



Prediction Intervals for Regression: Theory, Methods, and Applications

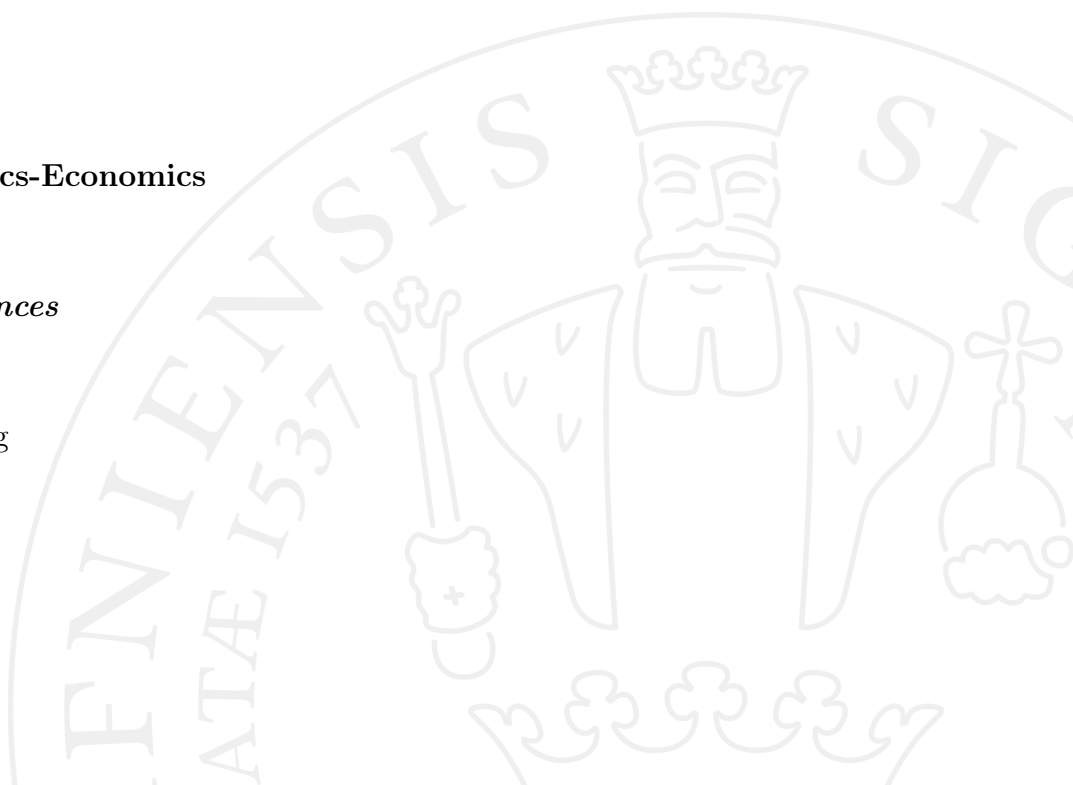
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Abstract

This thesis investigates the construction of prediction intervals for regression problems, with a particular focus on the trade-offs between validity, efficiency, and local adaptivity. The analysis considers classical Gaussian prediction intervals, quantile regression methods, and conformal prediction techniques, including split conformal prediction, Mondrian conformal prediction, and conformalized quantile regression. The methods are studied both theoretically and empirically through simulation experiments and a real-world case study of housing prices from the Capital Region of Denmark. The results demonstrate that no single approach consistently outperforms the alternatives across all experimental settings. Conformal methods offer robust distribution-free finite-sample coverage guarantees, while methods based on conditional quantile estimation may achieve greater efficiency and local adaptivity when their modeling assumptions are satisfied. The empirical studies further illustrate how the relative strengths of different approaches depend on the underlying data-generating process, sample size, and degree of model misspecification. Overall, the thesis highlights the fundamental trade-offs underlying modern prediction interval construction and demonstrates the challenges of simultaneously achieving validity, efficiency, and local adaptivity in complex predictive settings.

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1 Introduction

Prediction problems arise in a wide range of scientific and practical applications, including medicine, finance, climate science, energy forecasting, and real estate valuation (James et al., 2013; Hastie et al., 2009). In many of these settings, point predictions alone are insufficient, since practical decision-making additionally depends on understanding the uncertainty associated with future observations. In housing markets, for example, uncertainty quantification plays a central role in mortgage lending, investment analysis, insurance valuation, and risk management (Hjort et al., 2024a; Besien, 2026). Similarly, in medical prognosis, weather forecasting, and financial risk assessment, decision-makers are often less interested in a single predicted value than in a reliable range of plausible outcomes (Balasubramanian et al., 2014). Consequently, the construction of prediction intervals has become an increasingly important problem in modern statistical learning and uncertainty quantification (Angelopoulos et al., 2025).

A prediction interval aims to quantify uncertainty about a future observation by constructing an interval expected to contain the unknown response with high probability. This differs fundamentally from the classical notion of confidence intervals. While confidence intervals quantify uncertainty about an unknown population parameter, such as a regression coefficient or population mean, prediction intervals quantify uncertainty about future realizations of the response variable itself (Pever and Kononenko, 2015; James et al., 2013). For example, in a housing price setting, a confidence interval may quantify uncertainty about the average effect of floor area on housing prices, whereas a prediction interval quantifies the uncertainty associated with the price of a specific future property transaction. Since future observations contain both parameter uncertainty and irreducible noise, prediction intervals are typically wider than confidence intervals (James et al., 2013). From a practical perspective, prediction intervals are therefore often more directly relevant for decision-making and risk assessment (Hjort et al., 2024a).

Constructing prediction intervals that provide useful uncertainty quantification remains challenging, as several desirable properties must be balanced simultaneously. First, the intervals should be valid, meaning that the true future response is contained within the interval with the desired probability. Second, the intervals should be efficient, meaning sufficiently narrow to provide informative uncertainty quantification. Third, the intervals should ideally adapt to local variation in predictive uncertainty across the covariate space, such as heteroskedasticity, asymmetry, or nonlinear structure in the conditional distribution (Lei et al., 2018; Angelopoulos et al., 2025).

Classical prediction intervals based on linear regression provide exact finite-sample validity under strong parametric assumptions, including linearity, homoskedastic Gaussian errors, and correct model specification (Montgomery et al., 2012). However, such assumptions are frequently violated in practice, particularly in the presence of nonlinear relationships, heteroskedasticity, or distributional asymmetry (Lei et al., 2018). These limitations have motivated the development of more flexible prediction interval methods. Quantile regression (Koenker and Bassett, 1978) estimates conditional quantiles directly and can therefore be

used to construct prediction intervals that naturally accommodate heteroskedasticity and asymmetric conditional distributions. More recently, machine-learning-based approaches such as quantile regression forests (Meinshausen, 2006) have extended these ideas to nonlinear settings. Although these methods provide greater flexibility and local adaptivity, their coverage properties depend on the quality of the estimated conditional quantiles, and they generally do not provide finite-sample validity guarantees (Meinshausen, 2006). In parallel, conformal prediction has emerged as a general framework for constructing prediction intervals with finite-sample distribution-free validity guarantees under minimal assumptions (Vovk et al., 2005). Conformal prediction may be combined with a wide range of underlying prediction algorithms, including both classical statistical models and modern machine learning methods. Under the sole assumption of exchangeability, conformal methods guarantee valid marginal coverage regardless of the correctness of the underlying prediction model (Lei et al., 2018; Angelopoulos et al., 2025). Due to these attractive theoretical properties, conformal prediction has received a lot of attention in recent years across a broad range of applications, including image classification, medicine, finance, genomics, and risk assessment (Balasubramanian et al., 2014; Angelopoulos et al., 2025). However, these guarantees are marginal and do not necessarily translate into accurate coverage across all regions of the covariate space. In particular, the conformal intervals may exhibit conditional miscoverage in heteroskedastic settings (Barber et al., 2021; Lei and Wasserman, 2014). As a result, a large and rapidly growing literature has investigated the trade-off between validity, efficiency, and adaptivity in prediction interval construction.

This thesis contributes to this literature through a comprehensive theoretical and empirical comparison of several widely used prediction interval methods. In particular, we study classical Gaussian prediction intervals, residual-based methods, quantile regression, split conformal prediction, Mondrian conformal prediction, and conformalized quantile regression. The methods are compared both theoretically and empirically with respect to validity, efficiency, local adaptivity, and robustness under model misspecification. The analysis combines three complementary perspectives. First, the thesis develops the theoretical foundations underlying the considered methods, including their assumptions, validity properties, and theoretical guarantees. Second, comprehensive simulation experiments investigate the practical behaviour of the methods under various forms of model and noise misspecification, including asymmetric noise, nonlinear regression structure, and heteroskedasticity. Finally, the methods are evaluated in an applied case study using highly heterogeneous housing price data from the Capital Region of Denmark. Overall, the results establish the fundamental statistical trade-offs underlying modern prediction interval construction and clarify the practical challenges of uncertainty quantification in complex predictive settings.

2 Prediction Intervals

In predictive regression settings, the goal is to predict an unknown response variable based on observed features. While point predictions provide a single estimate of the response, they do not quantify the uncertainty associated with the prediction. For continuous responses, prediction intervals provide a natural framework for uncertainty quantification by specifying a range of plausible values for the future outcome (James et al., 2013).

Different approaches to constructing prediction intervals rely on different assumptions and provide different types of validity guarantees. Classical prediction intervals rely on strong parametric assumptions, including linearity, Gaussian homoskedastic errors, and correct model specification, but in return provide exact finite-sample validity. Quantile regression methods relax many of these assumptions by modeling conditional quantiles directly and allowing for heterogeneous uncertainty across the covariate space. However, this increased flexibility comes at the cost of weaker, asymptotic, and model-dependent guarantees. Finally, conformal prediction methods provide distribution-free finite-sample marginal validity guarantees while requiring only exchangeability of the observations. However, these guarantees are marginal and do not necessarily translate into accurate coverage across all regions of the covariate space (Vovk et al., 2005; Meinshausen, 2006; Barber et al., 2021).

In this section, we first introduce the general regression setup and the evaluation criteria used throughout the thesis. We then present the main approaches to constructing prediction intervals, beginning with classical prediction intervals, followed by quantile regression methods and conformal prediction.

2.1 Preliminaries

We consider a random-design setting in which

$$(X_i, Y_i) \in \mathcal{X} \times \mathcal{Y}, \quad i = 1, \dots, n+1,$$

are random variables. Here, $X_i \in \mathcal{X} \subseteq \mathbb{R}^p$ denotes a feature vector and $Y_i \in \mathcal{Y} \subseteq \mathbb{R}$ denotes the corresponding response variable.

Given observed data $(X_1, Y_1), \dots, (X_n, Y_n)$, the goal is to predict the unobserved response Y_{n+1} corresponding to a new feature vector X_{n+1} . Let $\hat{\mu} : \mathcal{X} \rightarrow \mathcal{Y}$ denote a prediction model that produces a point prediction $\hat{\mu}(X_{n+1})$ for the response corresponding to the feature vector X_{n+1} .

Depending on the method under consideration, the prediction model may estimate different characteristics of the conditional distribution of $Y \mid X$, such as the conditional mean or conditional quantiles. Throughout this thesis, we primarily consider two underlying prediction models: a parametric linear regression model and a nonparametric random forest model.

Linear regression (Montgomery et al., 2012) assumes that the conditional mean of the response is linear in the covariates,

$$Y_i = X_i^\top \beta + \varepsilon_i,$$

where $\beta \in \mathbb{R}^p$ is an unknown parameter vector and ε_i denotes a random error term satisfying $\mathbb{E}[\varepsilon_i | X_i] = 0$. The parameter vector is estimated using the ordinary least squares estimator,

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n (Y_i - X_i^\top \beta)^2,$$

and predictions are then obtained through $\hat{\mu}(X_{n+1}) = X_{n+1}^\top \hat{\beta}$.

In contrast, random forests (Breiman, 2001) are nonparametric ensemble methods based on bootstrap aggregation and recursive partitioning, allowing for flexible modeling of complex relationships, including nonlinear effects. Rather than fitting a single decision tree, the method aggregates predictions from many randomized trees in order to reduce variance and improve predictive performance. The random forest procedure is summarized in Algorithm 2.1 (Hastie et al., 2009).

Algorithm 2.1: Random forest regression

1. **Input:** Training data

$$(X_1, Y_1), \dots, (X_n, Y_n),$$

number of trees B , and number of candidate features m .

2. **Bootstrap sampling:** For each tree $b = 1, \dots, B$, draw a bootstrap sample from the training data.

3. **Tree construction:** Grow a decision tree on each bootstrap sample:

- At each split, randomly select m candidate features.
- Choose the split minimizing the residual sum of squares.
- Grow the tree recursively until a stopping criterion is met.

4. **Prediction:** For a new feature vector X_{n+1} , compute the prediction from each tree

$$\hat{\mu}_b(X_{n+1}), \quad b = 1, \dots, B.$$

5. **Aggregation:** Average the tree predictions:

$$\hat{\mu}(X_{n+1}) = \frac{1}{B} \sum_{b=1}^B \hat{\mu}_b(X_{n+1}).$$

6. **Output:** Return

$$\hat{\mu}(X_{n+1}).$$

Having introduced the prediction models considered throughout the thesis, we now turn to the construction and evaluation of prediction intervals. To quantify uncertainty in the point prediction $\hat{\mu}(X_{n+1})$, we construct a prediction interval $C(X_{n+1}) \subseteq \mathcal{Y}$, intended to contain the future response Y_{n+1} with high probability. The quality of a prediction interval is typically assessed through two fundamental criteria: validity and efficiency. Validity concerns whether the interval attains at least the desired coverage, while efficiency measures how informative the interval is through its width (Lei et al., 2013).

Definition 2.1: Validity and coverage

A prediction interval

$$C(X_{n+1}) \subseteq \mathcal{Y}$$

is said to be valid at level $1 - \alpha$ if

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha.$$

Throughout the thesis, the phrase "at level $(1 - \alpha)$ " is often omitted when the target coverage level is clear from the context.

The quantity $\mathbb{P}(Y_{n+1} \in C(X_{n+1}))$ is referred to as the coverage of the prediction interval.

While validity concerns the coverage properties of the prediction interval, it does not capture how informative the interval is. In practice, excessively wide intervals may achieve high coverage while providing little practical value. This motivates the notion of efficiency.

Definition 2.2: Efficiency

For a prediction interval

$$C(X_{n+1}) = [L(X_{n+1}), U(X_{n+1})],$$

the measure of efficiency is defined as the length of the interval

$$\mathcal{L} = U(X_{n+1}) - L(X_{n+1}).$$

Prediction intervals with shorter length are considered more efficient, provided the desired coverage level is maintained.

In practice, there is often a trade-off between validity and efficiency (Lei et al., 2018). Methods producing very narrow intervals may fail to achieve the desired coverage level, whereas overly conservative methods may yield unnecessarily wide prediction intervals.

We now introduce the main classes of prediction interval methods considered throughout this thesis and discuss their respective approaches to balancing validity, efficiency, and the ability to adapt to local variation in predictive uncertainty.

2.2 Classical Prediction Intervals

We first consider classical prediction intervals based on the linear regression model introduced in Section 2.1. In addition to the linearity assumption, classical prediction intervals require the error terms to satisfy

$$\varepsilon_i \perp\!\!\!\perp X_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2).$$

Under these assumptions, the conditional distribution of the response given the covariates is Gaussian,

$$Y_i \mid X_i \sim \mathcal{N}(X_i^\top \beta, \sigma^2).$$

Consequently, the prediction error

$$Y_{n+1} - \hat{\mu}(X_{n+1})$$

admits an explicit distribution, which forms the basis for constructing classical prediction intervals.

Proposition 2.1: Distribution of the prediction error

Under the classical linear regression model

$$Y_i = X_i^\top \beta + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \quad \varepsilon_i \perp\!\!\!\perp X_i,$$

the prediction error satisfies

$$Y_{n+1} - \hat{\mu}(X_{n+1}) \mid X \sim \mathcal{N}\left(0, \sigma^2 \left(1 + X_{n+1}^\top (X^\top X)^{-1} X_{n+1}\right)\right).$$

Proof. Using the linear model,

$$Y_{n+1} = X_{n+1}^\top \beta + \varepsilon_{n+1},$$

the prediction error can be written as

$$Y_{n+1} - \hat{\mu}(X_{n+1}) = X_{n+1}^\top (\beta - \hat{\beta}) + \varepsilon_{n+1}.$$

Conditionally on X , the ordinary least squares estimator satisfies

$$\hat{\beta} \mid X \sim \mathcal{N}\left(\beta, \sigma^2 (X^\top X)^{-1}\right).$$

Since $\hat{\beta}$ and ε_{n+1} are Gaussian and independent, the prediction error is Gaussian as well. As the ordinary least squares estimator is unbiased conditionally on X , it follows that

$$\begin{aligned} \mathbb{E}(Y_{n+1} - \hat{\mu}(X_{n+1}) \mid X) &= \mathbb{E}(X_{n+1}^\top (\beta - \hat{\beta}) \mid X) + \mathbb{E}(\varepsilon_{n+1}) \\ &= X_{n+1}^\top (\beta - \mathbb{E}(\hat{\beta} \mid X)) + 0 \\ &= 0. \end{aligned}$$

Moreover,

$$\begin{aligned}
\text{Var}(Y_{n+1} - \hat{\mu}(X_{n+1}) \mid X) &= \text{Var}(X_{n+1}^\top(\beta - \hat{\beta}) \mid X) + \text{Var}(\varepsilon_{n+1}) \\
&= X_{n+1}^\top \text{Var}(\hat{\beta} \mid X) X_{n+1} + \sigma^2 \\
&= \sigma^2 (1 + X_{n+1}^\top (X^\top X)^{-1} X_{n+1}).
\end{aligned}$$

□

If the noise variance σ^2 is known, the standardized prediction error satisfies

$$\left. \frac{Y_{n+1} - \hat{\mu}(X_{n+1})}{\sigma \sqrt{1 + X_{n+1}^\top (X^\top X)^{-1} X_{n+1}}} \right| X \sim \mathcal{N}(0, 1).$$

Consequently, a $(1 - \alpha)$ -prediction interval is given by

$$\hat{\mu}(X_{n+1}) \pm z_{1-\alpha/2} \cdot \sigma \sqrt{1 + X_{n+1}^\top (X^\top X)^{-1} X_{n+1}},$$

where $z_{1-\alpha/2}$ denotes the $(1 - \alpha/2)$ -quantile of the standard normal distribution.

In practice, the variance σ^2 is typically unknown and must be estimated from the data. A standard estimator is

$$\hat{\sigma}^2 = \frac{1}{n - p} \|Y - X\hat{\beta}\|^2.$$

Under the Gaussian error assumption, the estimator $\hat{\sigma}^2$ satisfies

$$\frac{(n - p)\hat{\sigma}^2}{\sigma^2} \sim \chi_{n-p}^2,$$

and is independent of $\hat{\beta}$ (Montgomery et al., 2012). Combining these results, the standardized prediction error can be written as a ratio of a standard normal variable and the square root of an independent chi-squared variable divided by its degrees of freedom. Hence,

$$\left. \frac{Y_{n+1} - \hat{\mu}(X_{n+1})}{\hat{\sigma} \sqrt{1 + X_{n+1}^\top (X^\top X)^{-1} X_{n+1}}} \right| X \sim t_{n-p},$$

where t_{n-p} denotes the Student t -distribution with $n - p$ degrees of freedom (Montgomery et al., 2012).

Hence, a $(1 - \alpha)$ -prediction interval is given by

$$\hat{\mu}(X_{n+1}) \pm t_{n-p, 1-\alpha/2} \cdot \hat{\sigma} \sqrt{1 + X_{n+1}^\top (X^\top X)^{-1} X_{n+1}},$$

where $t_{n-p, 1-\alpha/2}$ denotes $(1 - \alpha/2)$ -quantile of the Student t -distribution with $n - p$ degrees of freedom.

Under the classical linear regression assumptions, the resulting prediction interval satisfies

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X) = 1 - \alpha.$$

Thus, the interval achieves exact finite-sample coverage conditionally on the observed covariates (Montgomery et al., 2012). However, this guarantee relies on strong assumptions, including Gaussian errors, homoskedasticity, independence, and correct model specification. In practice, such assumptions are often violated, motivating the development of prediction interval methods with weaker assumptions.

2.3 Quantile Regression

While the classical prediction intervals considered in Section 2.2 rely on strong parametric assumptions, quantile regression (QR) (Koenker and Bassett, 1978) relaxes these assumptions by modeling conditional quantiles directly rather than the conditional mean $\mathbb{E}[Y \mid X = x]$. Specifically, QR models the conditional τ -quantile

$$q_\tau(x) = \inf\{y : F(y \mid x) \geq \tau\},$$

thereby allowing the resulting prediction intervals to adapt to heteroskedasticity and local variation in the conditional distribution. However, the method still relies on assumptions regarding the estimation of the conditional quantiles and provides only asymptotic and model-dependent guarantees under standard regularity conditions (Koenker and Bassett, 1978).

For $\tau \in (0, 1)$, define the pinball loss

$$\rho_\tau(y - q) = \begin{cases} \tau|y - q|, & y > q, \\ (1 - \tau)|y - q|, & y \leq q. \end{cases}$$

While the conditional mean minimizes the expected squared error loss, the conditional τ -quantile satisfies

$$q_\tau(x) = \arg \min_{q \in \mathbb{R}} \mathbb{E}[\rho_\tau(Y - q) \mid X = x].$$

Thus, conditional quantiles may be characterized as minimizers of an asymmetric loss function. Classical quantile regression estimates these conditional quantiles by minimizing the empirical pinball loss, while more flexible machine learning approaches may estimate the conditional quantiles through alternative procedures.

2.3.1 Linear Quantile Regression

Linear quantile regression (Koenker and Bassett, 1978) assumes that the conditional τ -quantile is linear in the covariates,

$$q_\tau(x) = x^\top \beta_\tau,$$

and estimates the parameter vector through

$$\hat{\beta}_\tau = \arg \min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n \rho_\tau(Y_i - X_i^\top \beta).$$

A prediction interval at level $1 - \alpha$ is constructed by estimating the lower and upper conditional quantiles corresponding to $\tau = \alpha/2$ and $\tau = 1 - \alpha/2$, yielding the interval

$$\left[X_{n+1}^\top \hat{\beta}_{\alpha/2}, X_{n+1}^\top \hat{\beta}_{1-\alpha/2} \right].$$

Unlike the Gaussian interval in Section 2.2, this interval is asymmetric and adapts to the shape of the conditional distribution. However, the assumed linear quantile structure may be misspecified in practice, particularly in the presence of nonlinear relationships (Meinshausen, 2006; Koenker and Bassett, 1978).

2.3.2 Quantile Regression Forests

To obtain a more flexible estimator of the conditional quantiles, we replace the linear model in 2.3.1 with a random forest prediction model introduced in Section 2.1. A standard random forest estimates the conditional mean by averaging the responses within each terminal node. Quantile regression forests (Meinshausen, 2006) extend this idea by retaining all training responses in each terminal node rather than only their average. This allows the full conditional distribution of the response to be estimated locally around the test point X_{n+1} .

The quantile regression forests procedure is summarized in Algorithm 2.2 (Meinshausen, 2006).

Algorithm 2.2: Quantile Random Forest

1. **Input:** Training data $(X_1, Y_1), \dots, (X_n, Y_n)$,
number of trees B , and target quantile level $\tau \in (0, 1)$.
2. **Forest construction:** Fit a random forest consisting of B regression trees using bootstrap samples of the training data.
3. **Terminal node storage:** For each tree, store all response values Y_i contained in every terminal node.
4. **Weight construction:** For a new feature vector X_{n+1} , assign weights to the training observations according to how frequently they occur in the same terminal nodes as X_{n+1} .
5. **Conditional distribution estimation:** Construct the weighted empirical distribution function

$$\hat{F}(y \mid X_{n+1}).$$

6. **Quantile estimation:** Estimate the conditional quantile

$$\hat{q}_\tau(X_{n+1}) = \inf \left\{ y : \hat{F}(y \mid X_{n+1}) \geq \tau \right\}.$$

7. **Output:** Return

$$\hat{q}_\tau(X_{n+1}).$$

Algorithm 2.2 summarizes the quantile regression forest procedure operationally. We next formalize the corresponding adaptive weight construction underlying the conditional quantile estimator and the resulting prediction interval.

Formally, let $T_b(x)$ denote the terminal node in tree b containing the point x , and let $|T_b(x)|$ denote the number of training observations contained in that node. The quantile regression forest (Meinshausen, 2006) assigns weights

$$w_i(x) = \frac{1}{B} \sum_{b=1}^B \frac{\mathbb{1}\{X_i \in T_b(x)\}}{|T_b(x)|},$$

which measure how frequently observation i appears in the same terminal node as x across the forest.

Using these weights, the conditional distribution function is estimated by

$$\hat{F}(y \mid x) = \sum_{i=1}^n w_i(x) \mathbb{1}\{Y_i \leq y\},$$

yielding a weighted empirical distribution of the training responses localized around x . The conditional quantile estimate is then obtained through

$$\hat{q}_\tau(x) = \inf \left\{ y : \hat{F}(y \mid x) \geq \tau \right\}.$$

A $(1 - \alpha)$ -prediction interval is constructed as

$$\left[\hat{q}_{\alpha/2}(X_{n+1}), \hat{q}_{1-\alpha/2}(X_{n+1}) \right].$$

Unlike linear quantile regression, quantile regression forest does not impose a parametric form on the conditional quantiles. Consequently, the interval width

$$\hat{q}_{1-\alpha/2}(x) - \hat{q}_{\alpha/2}(x)$$

may vary flexibly across the covariate space, allowing the method to adapt naturally to heteroskedasticity and nonlinear structure in the data. Although quantile regression forest intervals provide greater flexibility and local adaptivity, their coverage properties depend on the quality of the estimated conditional quantiles. Consequently, they do not provide finite-sample validity guarantees. Instead, Meinshausen (2006) established consistency of the estimated conditional quantiles under suitable regularity conditions, implying that the corresponding prediction intervals are justified only asymptotically.

2.4 Conformal Prediction

We now turn to conformal prediction (Vovk et al., 2005; Gammerman et al., 1998) which provides a general framework for constructing prediction intervals with finite-sample validity under minimal assumptions. In the setting introduced in Section 2.1, the goal remains to construct a prediction interval

$$C(X_{n+1}) \subseteq \mathcal{Y}$$

such that

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha.$$

A key advantage of conformal prediction is that it may be combined with a wide range of prediction algorithms, including both parametric and nonparametric models, while still retaining valid marginal coverage guarantees without requiring distributional assumptions on the data-generating process (Lei et al., 2018).

Define

$$Z_i = (X_i, Y_i) \in \mathcal{Z} := \mathcal{X} \times \mathcal{Y}, \quad i = 1, \dots, n+1.$$

The validity of conformal prediction relies on the assumption that the observations $Z_1, \dots, Z_{n+1}$ are exchangeable. The notion of exchangeability formalizes the idea that the ordering of the observations should not affect their joint distribution (Angelopoulos et al., 2025; Shafer and Vovk, 2024).

Definition 2.3: Exchangeability

Let $Z_1, \dots, Z_n$ be random variables. The vector $(Z_1, \dots, Z_n)$ is said to be exchangeable if

$$(Z_1, \dots, Z_n) \stackrel{d}{=} (Z_{\sigma(1)}, \dots, Z_{\sigma(n)})$$

for every permutation σ .

Exchangeability is weaker than independent and identically distributed sampling, but still implies that the ordering of the observations carries no information.

The central idea of conformal prediction is to compare a candidate observation to previously observed data using a nonconformity score. Prediction intervals are then constructed by identifying response values that appear sufficiently consistent with the observed data.

Definition 2.4: Nonconformity score

A nonconformity score is a function

$$s : \mathcal{Z} \times \mathcal{Z}^{n+1} \rightarrow \mathbb{R}$$

that measures how atypical an observation is relative to a dataset. Given an observation Z_i and a dataset $\mathcal{D}$, the score

$$S_i = s(Z_i; \mathcal{D})$$

quantifies how poorly the observation conforms to the remaining data.

Intuitively, a nonconformity score measures how unusual an observation (x, y) is relative to the training data. Larger values indicate lower conformity. A common choice is the residual score

$$s((x, y); \mathcal{D}) = |y - \hat{\mu}(x)|,$$

where $\hat{\mu}$ is estimated from the dataset $\mathcal{D}$. Here, larger prediction errors correspond to larger nonconformity scores.

The validity of conformal prediction follows from a simple rank argument. Intuitively, exchangeability implies that a future observation should not systematically appear more or less atypical than the observed data. The following proposition formalizes this observation by showing that the rank of an exchangeable observation is uniformly distributed.

Proposition 2.2: Uniform rank under exchangeability

Let

$$S_1, \dots, S_{n+1}$$

be exchangeable and almost surely distinct random variables. Define the rank of S_{n+1} among the $n + 1$ observations by

$$\text{rank}(S_{n+1}) = 1 + \sum_{i=1}^n \mathbb{1}\{S_i \leq S_{n+1}\}.$$

Then,

$$\text{rank}(S_{n+1}) \sim \text{Uniform}\{1, \dots, n + 1\}.$$

Proof. By exchangeability, all $(n + 1)!$ orderings of

$$S_1, \dots, S_{n+1}$$

are equally likely. Hence, the rank of S_{n+1} among the $n + 1$ observations is uniformly distributed on

$$\{1, \dots, n + 1\}.$$

□

The assumption of almost sure distinctness is imposed to avoid ties. In the presence of ties, exact uniformity can be recovered by introducing a random tie-breaking mechanism.

2.4.1 Full Conformal Prediction

We first describe the full conformal prediction procedure for prediction intervals in regression (Lei et al., 2018). The key idea is to evaluate how well a candidate response value y conforms to the observed data by comparing its nonconformity score to those of the training observations.

Define

$$Z_i = (X_i, Y_i),$$

and assume that

$$Z_1, \dots, Z_{n+1}$$

are exchangeable. Let $s : \mathcal{Z} \times \mathcal{Z}^{n+1} \rightarrow \mathbb{R}$ be a symmetric nonconformity score function, meaning that the resulting scores are invariant under permutations of the observations. The full conformal prediction procedure is summarized in Algorithm 2.3 (Angelopoulos et al., 2025).

Algorithm 2.3: Full conformal prediction

1. **Input:** Data

$$(X_1, Y_1), \dots, (X_n, Y_n),$$

a test point X_{n+1} , a target coverage level $1 - \alpha$, and a symmetric nonconformity score function s .

2. **Augmentation:** For each candidate response value $y \in \mathcal{Y}$, construct the augmented dataset

$$\mathcal{D}_y = \{(X_1, Y_1), \dots, (X_n, Y_n), (X_{n+1}, y)\}.$$

3. **Score computation:** Compute the nonconformity scores

$$S_i^y = s((X_i, Y_i); \mathcal{D}_y), \quad i = 1, \dots, n,$$

and

$$S_{n+1}^y = s((X_{n+1}, y); \mathcal{D}_y).$$

4. **Quantile estimation:** Let $\hat{q}_{1-\alpha}^y$ denote the empirical $(1 - \alpha)$ -quantile of $\{S_1^y, \dots, S_n^y\}$, defined as

$$\hat{q}_{1-\alpha}^y = S_{(k^*)}^y, \quad k^* = \lceil (n+1)(1-\alpha) \rceil.$$

5. **Output:** Return the prediction interval

$$C(X_{n+1}) = \{y \in \mathcal{Y} : S_{n+1}^y \leq \hat{q}_{1-\alpha}^y\}.$$

To understand why the full conformal procedure achieves valid coverage, we study the rank of the test score among the augmented scores. The following proposition establishes the resulting finite-sample validity guarantee.

Proposition 2.3: Validity of full conformal prediction

For a candidate response value y , define

$$\pi(y) = \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbb{1}\{S_i^y \leq S_{n+1}^y\}.$$

Then, $1 - \pi(y)$ may be interpreted as a conformal p-value for testing the null hypothesis

$$H_0 : Y_{n+1} = y.$$

Moreover,

$$\mathbb{P}(\pi(Y_{n+1}) \leq 1 - \alpha) \geq 1 - \alpha.$$

Consequently, the prediction interval

$$C(X_{n+1}) = \{y \in \mathbb{R} : \pi(y) \leq 1 - \alpha\}.$$

satisfies

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha.$$

Proof. The quantity $\pi(y)$ corresponds to the normalized rank of the test score S_{n+1}^y among the augmented scores. When $y = Y_{n+1}$, the augmented dataset coincides with the full sample. Since the observations are exchangeable and the score function is symmetric, the scores $S_1^{Y_{n+1}}, \dots, S_{n+1}^{Y_{n+1}}$ are exchangeable. Consequently, by Proposition 2.2,

$$(n+1)\pi(Y_{n+1}) \sim \text{Uniform}\{1, \dots, n+1\}.$$

Hence,

$$P((n+1)\pi(Y_{n+1}) \leq \lceil (1-\alpha)(n+1) \rceil) = \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} \geq 1 - \alpha.$$

This motivates defining the conformal prediction interval as

$$C(X_{n+1}) = \{y \in \mathbb{R} : (n+1)\pi(y) \leq \lceil (1-\alpha)(n+1) \rceil\}.$$

By construction,

$$P(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha.$$

□

While the rank-based formulation is useful for understanding the finite-sample validity of conformal prediction, the procedure is often implemented through an equivalent quantile representation as in Algorithm 2.3. Hence, we rewrite the conformal prediction interval in terms of empirical score quantiles, yielding a more convenient characterization of the full conformal procedure.

Since

$$\pi(y) = \frac{\text{rank}(S_{n+1}^y)}{n+1},$$

the prediction interval may equivalently be written in quantile form. Define

$$k^* = \lceil (n+1)(1-\alpha) \rceil,$$

and let

$$\hat{q}_{1-\alpha}^y = S_{(k^*)}^y$$

denote the empirical $(1-\alpha)$ -quantile of the augmented scores

$$\{S_1^y, \dots, S_n^y\}.$$

Then,

$$C(X_{n+1}) = \{y \in \mathbb{R} : S_{n+1}^y \leq \hat{q}_{1-\alpha}^y\}.$$

Importantly, this validity guarantee holds exactly in finite samples and does not rely on asymptotic approximations or parametric distributional assumptions beyond exchangeability.

A major drawback of full conformal prediction is its computational cost. Since the prediction model must generally be refitted for every candidate response value y , the procedure can become computationally demanding, particularly for complex prediction models.

2.4.2 Split Conformal Prediction

An alternative to the full (transductive) conformal prediction method is split conformal prediction (Papadopoulos et al., 2002; Lei et al., 2018), which separates model fitting and calibration through sample splitting. This method is also known as inductive conformal prediction (ICP) in the literature, reflecting the fact that the prediction rule is learned on one part of the data and then applied to another.

The data (X_i, Y_i) , $i = 1, \dots, n$, are divided into two disjoint subsets I_1 and I_2 . The first subset is used to fit the prediction model, while the second is used to construct the prediction interval. For simplicity, we assume that n is even and consider a balanced split, i.e., $|I_1| = |I_2| = n/2$. More generally, one may use an unbalanced split with $|I_1| = \rho n$ and $|I_2| = (1-\rho)n$ for some $\rho \in (0, 1)$. The choice of ρ reflects a trade-off between the accuracy of the fitted model (larger I_1) and the precision of the calibration step (larger I_2) (Lei et al., 2018). In this thesis, we restrict attention to the balanced case. Using the observations indexed by I_1 , a prediction model $\hat{\mu}$ is fitted, while the observations in I_2 are used to calibrate the resulting prediction interval. The split conformal prediction procedure is summarized in Algorithm 2.4.

Algorithm 2.4: Split conformal prediction

1. **Input:** Data $(X_1, Y_1), \dots, (X_n, Y_n)$, a test point X_{n+1} , a target coverage level $1 - \alpha$, and a nonconformity score function s .
2. **Split the data:** Partition the indices $\{1, \dots, n\}$ into two disjoint sets I_1 and I_2 , with $|I_1| = |I_2| = n/2$.
3. **Model fitting:** Fit a regression estimator $\hat{\mu}$ using the data $\{(X_i, Y_i) : i \in I_1\}$.
4. **Calibration:** Compute the nonconformity scores on the calibration set

$$S_i = s((X_i, Y_i)), \quad i \in I_2.$$

5. **Quantile estimation:** Let $\hat{q}_{1-\alpha}$ denote the empirical $(1 - \alpha)$ -quantile of $\{S_i : i \in I_2\}$, defined as

$$\hat{q}_{1-\alpha} = S_{(k^*)}, \quad k^* = \lceil (|I_2| + 1)(1 - \alpha) \rceil.$$

6. **Output:** Return the prediction interval

$$C_{\text{CP}}(X_{n+1}) = \{y \in \mathcal{Y} : s((X_{n+1}, y)) \leq \hat{q}_{1-\alpha}\}.$$

Unlike full conformal prediction, where the nonconformity score depends on the augmented dataset through repeated model refitting, split conformal prediction fits the prediction model only once. Consequently, the nonconformity score may be viewed as a function of a single observation,

$$s : \mathcal{Z} \rightarrow \mathbb{R},$$

with the fitted prediction model treated as fixed throughout the calibration step. Since no repeated model refitting is required for candidate response values y , the computational complexity is essentially that of a single model fit, making split conformal prediction more scalable than full conformal prediction.

Throughout this thesis, we exclusively consider split conformal methods and their extensions. Consequently, unless otherwise stated, the terms conformal prediction and conformal inference will refer specifically to the split conformal framework.

Furthermore, all conformal methods considered in the thesis are based on the absolute residual score

$$s((X_i, Y_i)) = |Y_i - \hat{\mu}(X_i)|.$$

With this choice of nonconformity score, the split conformal prediction interval takes the form

$$C_{\text{CP}}(x_{n+1}) = [\hat{\mu}(x_{n+1}) \pm \hat{q}_{1-\alpha}].$$

2.4.3 Validity of Split Conformal Prediction

The validity of split conformal prediction can be studied through different notions of coverage. The fundamental guarantee provided by split conformal prediction is marginal validity, which holds distribution-free under the sole assumption of exchangeability.

However, marginal validity does not imply that the prediction intervals achieve the desired coverage conditionally on the covariates. This motivates stronger notions of validity, such as conditional and training-conditional validity, as well as localized conformal methods that aim to improve local coverage behavior.

Marginal validity

We begin by studying the marginal validity guarantee of split conformal prediction, formalized in Theorem 2.1 (Lei et al., 2018, Theorem 2.2)

Theorem 2.1: Validity of split conformal prediction

Assume that

$$(X_1, Y_1), \dots, (X_n, Y_n), (X_{n+1}, Y_{n+1})$$

are exchangeable.

Let $C_{\text{CP}}(X_{n+1})$ denote the split conformal prediction interval constructed using the absolute residual scores

$$R_i = |Y_i - \hat{\mu}(X_i)|, \quad i \in I_2,$$

where the prediction model $\hat{\mu}$ is fitted using only the training set I_1 .

Then,

$$\mathbb{P}(Y_{n+1} \in C_{\text{CP}}(X_{n+1})) \geq 1 - \alpha.$$

If, additionally, the absolute residuals have a continuous joint distribution, then

$$\mathbb{P}(Y_{n+1} \in C_{\text{CP}}(X_{n+1})) \leq 1 - \alpha + \frac{2}{n+2}.$$

Proof. Define the absolute test residual

$$R_{n+1} = |Y_{n+1} - \hat{\mu}(X_{n+1})|.$$

Since the prediction model $\hat{\mu}$ is fitted using only the observations in I_1 , and the observations are exchangeable, the absolute calibration residuals

$$R_i = |Y_i - \hat{\mu}(X_i)|, \quad i \in I_2,$$

together with R_{n+1} are exchangeable. Let

$$k^* = \lceil (n/2 + 1)(1 - \alpha) \rceil,$$

and let

$$\hat{q}_{1-\alpha} = R_{(k^*)}$$

denote the empirical $(1 - \alpha)$ -quantile of the absolute residuals $\{R_i : i \in I_2\}$. By construction,

$$Y_{n+1} \in C_{\text{CP}}(X_{n+1}) \iff R_{n+1} \leq \hat{q}_{1-\alpha}.$$

Equivalently,

$$Y_{n+1} \in C_{\text{CP}}(X_{n+1})$$

if and only if the rank of R_{n+1} among the $n/2 + 1$ absolute residuals is at most k^* .

By Proposition 2.2,

$$\text{rank}(R_{n+1}) \sim \text{Uniform}\{1, \dots, n/2 + 1\}.$$

Hence,

$$\mathbb{P}(Y_{n+1} \in C_{\text{CP}}(X_{n+1})) \geq \frac{\lceil (n/2 + 1)(1 - \alpha) \rceil}{n/2 + 1} \geq 1 - \alpha.$$

If the absolute residuals additionally have a continuous joint distribution, ties occur with probability zero. In this case, the previous inequality sharpens to

$$\mathbb{P}(Y_{n+1} \in C_{\text{CP}}(X_{n+1})) = \frac{\lceil (n/2 + 1)(1 - \alpha) \rceil}{n/2 + 1}.$$

The interval then excludes those values of y for which $|y - x_{n+1}^\top \hat{\beta}|$ ranks among the largest absolute residuals. Then,

$$\begin{aligned} \mathbb{P}(Y_{n+1} \notin C_{\text{CP}}(X_{n+1})) &= 1 - \frac{\lceil (n/2 + 1)(1 - \alpha) \rceil}{n/2 + 1} \\ &= \frac{(n/2 + 1) - \lceil (n/2 + 1)(1 - \alpha) \rceil}{n/2 + 1} \\ &\geq \frac{(n/2 + 1) - (n/2 + 1)(1 - \alpha) - 1}{n/2 + 1} \\ &= 1 - (1 - \alpha) - \frac{1}{n/2 + 1} \\ &= \alpha - \frac{2}{n + 2}, \end{aligned}$$

where the inequality follows from $\lceil (n/2 + 1)(1 - \alpha) \rceil \leq (n/2 + 1)(1 - \alpha) + 1$.

Hence,

$$\mathbb{P}(Y_{n+1} \in C_{\text{CP}}(X_{n+1})) \leq 1 - \alpha + \frac{2}{n + 2}.$$

Thus, split conformal prediction guarantees marginal coverage at least $1 - \alpha$, while the excess coverage above the nominal level is at most $\frac{2}{n+2}$ under the additional assumption that the absolute residuals have a continuous joint distribution (Lei et al., 2018; Tibshirani, 2023). $\square$

Conditional validity

A stronger notion of validity is conditional validity, which requires

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x) \geq 1 - \alpha, \quad x \in \mathcal{X}.$$

Unlike marginal validity, conditional validity requires the prediction interval to achieve the target coverage level at every covariate value individually.

Indeed, marginal coverage may be written as

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) = \mathbb{E}[\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1})].$$

Thus, marginal validity only guarantees that the conditional coverage averages to at least $1 - \alpha$ over the covariate distribution. Consequently, some regions of the covariate space may exhibit undercoverage, compensated by overcoverage elsewhere.

For split conformal prediction, the marginal validity guarantee relies on the exchangeability of the nonconformity scores. However, this exchangeability argument only holds marginally. Conditional on $X_{n+1} = x$, the distribution of the absolute test residual may differ from the absolute calibration residuals, particularly in heteroskedastic settings or under model misspecification. In continuous settings, there are typically no calibration observations exactly at a given covariate value x , making it impossible to localize the exchangeability argument to a single point in the covariate space. Consequently, exact conditional validity generally cannot be guaranteed (Lei and Wasserman, 2014).

Formally, let

$$\mathcal{N}(P) = \{x \in \mathcal{X} : \mathbb{P}(X = x) = 0\}$$

denote the set of non-atoms of the distribution of X , that is, the set of feature values that occur with zero probability individually. If X has a continuous distribution, then every point in the feature space is a non-atom.

The fundamental limitation of distribution-free prediction intervals was established by Lei and Wasserman (2014). They show that any prediction interval satisfying distribution-free conditional validity must necessarily have infinite length for all $x \in \mathcal{N}(P)$.

Although exact distribution-free conditional validity is unattainable in general, the result motivates the search for methods that better adapt to local variation in predictive uncertainty while retaining marginal validity.

Mondrian conformal prediction

One approach to improving local adaptivity is to localize the conformal calibration step, yielding prediction intervals that better reflect local variation in predictive uncertainty across the covariate space. Mondrian conformal prediction (Vovk et al., 2003) partitions the covariate space into subsets

$$\mathcal{X} = \bigcup_{k=1}^K \mathcal{X}_k,$$

and computes separate conformal quantiles within each region (Barber et al., 2021). The resulting prediction interval takes the form

$$C(x) = [\hat{\mu}(x) - \hat{q}_{1-\alpha}^{(k)}, \hat{\mu}(x) + \hat{q}_{1-\alpha}^{(k)}], \quad x \in \mathcal{X}_k.$$

A particularly important special case arises when the covariate space is discrete,

$$\mathcal{X} = \{x_1, \dots, x_K\}.$$

Here the partition is given by the individual covariate values (Angelopoulos et al., 2025). Since calibration observations satisfying $X_i = x_k$ are exchangeable with future observations satisfying $X_{n+1} = x_k$, the standard conformal rank argument applies within each group, yielding

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid X_{n+1} = x_k) \geq 1 - \alpha.$$

Mondrian conformal prediction allows the interval width to adapt across the covariate space and may improve local coverage properties. However, smaller local calibration sets increase the variability of the estimated quantiles, potentially leading to unstable prediction intervals.

Training-conditional validity

Although exact conditional validity is generally unattainable in a distribution-free setting, one may instead study the weaker notion of training-conditional coverage (Angelopoulos et al., 2025; Vovk, 2012),

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid \mathcal{D}_n),$$

which conditions only on the realized training and calibration dataset. Training-conditional coverage quantifies how much the realized coverage may fluctuate across different realizations of the dataset, even when the marginal guarantee holds exactly.

Marginal validity implies that

$$1 - \alpha \leq \mathbb{P}(Y_{n+1} \in C(X_{n+1})) = \mathbb{E}[\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid \mathcal{D}_n)].$$

Hence, the marginal coverage guarantee may be interpreted as a statement about the expected training-conditional coverage over all possible realizations of the training data.

Unlike the marginal validity guarantee, which only requires exchangeability, training-conditional validity relies on the stronger assumption that the observations are i.i.d. It has been shown by Angelopoulos et al. (2025); Elder (2016) and Marchal and Arbel (2017) that the training-conditional coverage stochastically dominates a

$$\text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1))$$

distribution. Furthermore, if the score distribution is continuous, such that ties occur with probability zero, and $(1 - \alpha)(n + 1)$ is an integer, the training-conditional coverage is exactly distributed as the Beta distribution

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid \mathcal{D}_n) \sim \text{Beta}((1 - \alpha)(n + 1), \alpha(n + 1)).$$

For a Beta distributed random variable $X \sim \text{Beta}(a, b)$, the mean satisfies

$$\mathbb{E}[X] = \frac{a}{a+b}.$$

Hence,

$$\mathbb{E}[\mathbb{P}(Y_{n+1} \in C(X_{n+1}) \mid \mathcal{D}_n)] = \frac{(1-\alpha)(n+1)}{(1-\alpha)(n+1) + \alpha(n+1)} = 1 - \alpha.$$

Thus, the expected training-conditional coverage coincides with the classical marginal conformal guarantee.

The variance of a Beta-distributed random variable is

$$\text{Var}(X) = \frac{ab}{(a+b)^2(a+b+1)}.$$

Substituting

$$a = (1-\alpha)(n+1), \quad b = \alpha(n+1),$$

yields

$$\text{Var}(X) = \frac{\alpha(1-\alpha)}{n+2}.$$

Consequently, the variability of the training-conditional coverage decreases at rate $1/n$. For small calibration sets, the realized coverage may deviate from the nominal level $1-\alpha$, whereas larger calibration sets yield increasingly stable prediction intervals.

Importantly, this Beta characterization relies on the specific structure of split conformal prediction. Since the pretrained nonconformity score function s is fixed independently of the calibration data, the calibration scores

$$S_i = s(X_i, Y_i)$$

are i.i.d. under the assumption that the data are i.i.d. This allows the training-conditional coverage to be represented through an order statistic, yielding the Beta distribution characterization. As shown by Angelopoulos et al. (2025) and Bian and Barber (2023), the Beta characterization is a special feature of split conformal prediction and generally does not hold for full conformal prediction. In the full conformal setting, the nonconformity scores depend jointly on the candidate response value and the augmented dataset,

$$S_i = s(X_i, Y_i; \mathcal{D}_{n+1}),$$

which induces dependence between the scores. Consequently, the order-statistic argument underlying the Beta characterization breaks down.

2.4.4 Efficiency of Split Conformal Prediction

While split conformal prediction guarantees finite-sample validity, an important question is whether this guarantee is achieved at the cost of excessively wide prediction intervals. To assess the efficiency of conformal prediction, it is therefore natural to compare its interval length to that of an idealized oracle procedure. Let $q_{\alpha/2}(x)$ and $q_{1-\alpha/2}(x)$ denote the conditional quantiles of $Y \mid X = x$. The oracle prediction interval is defined as

$$C^*(x) = [q_{\alpha/2}(x), q_{1-\alpha/2}(x)],$$

which attains exact conditional validity and has the smallest possible length among all valid prediction intervals.

Consider the special case where the observations are i.i.d. and

$$Y = \mu(X) + \varepsilon,$$

where $\mu(x) = \mathbb{E}[Y \mid X = x]$, and ε is independent of X and symmetric around zero. Let $q_{1-\alpha}$ denote the $(1 - \alpha)$ -quantile of $|\varepsilon|$. The oracle interval then simplifies to

$$C^*(x) = [\mu(x) - q_{1-\alpha}, \mu(x) + q_{1-\alpha}],$$

with constant length

$$\nu^* = 2q_{1-\alpha}.$$

By comparison, split conformal prediction constructs intervals of the form

$$C_{\text{CP}}(x) = [\hat{\mu}(x) - \hat{q}_{1-\alpha}, \hat{\mu}(x) + \hat{q}_{1-\alpha}],$$

where $\hat{\mu}$ is an estimator of the regression function and $\hat{q}_{1-\alpha}$ is the empirical conformal quantile computed from the absolute calibration residuals. As a result, the conformal interval length is

$$\nu_{\text{CP}} = 2\hat{q}_{1-\alpha}.$$

The key observation is that if $\hat{\mu}(x)$ approximates $\mu(x)$ accurately, then the absolute calibration residuals $|Y_i - \hat{\mu}(X_i)|$ approximately follow the same distribution as the absolute noise $|\varepsilon|$. Consequently, the conformal quantile $\hat{q}_{1-\alpha}$ approaches the oracle quantile $q_{1-\alpha}$, implying that the conformal interval becomes nearly as short as the oracle interval.

More formally, Lei et al. (2018) have shown that if

$$\mathbb{E}[(\hat{\mu}(X) - \mu(X))^2] \rightarrow 0,$$

then the excess length of the conformal interval vanishes asymptotically, and

$$\nu_{\text{CP}} \xrightarrow{\mathbb{P}} 2q_{1-\alpha}.$$

Thus, the efficiency of split conformal prediction depends directly on the quality of the underlying regression estimator. Under suitable consistency assumptions, the conformal interval length converges to the oracle length, implying that split conformal prediction achieves

asymptotically optimal interval length while retaining exact finite-sample marginal validity.

The homoskedastic setting considered above represents a particularly simple case. If the noise is heteroskedastic or asymmetric, the oracle interval will generally depend on x , reflecting local variation in predictive uncertainty. In such settings, a constant interval width may no longer be optimal, even though split conformal prediction continues to provide marginal validity.

2.5 Summary

The prediction interval methods considered in this chapter illustrate different trade-offs between assumptions, validity guarantees, and adaptivity to the underlying data-generating distribution. Classical prediction intervals derived from linear regression theory provide exact finite-sample validity conditional on the observed covariates. However, these guarantees rely on strong parametric assumptions, including linearity, Gaussian homoskedastic errors, and correct model specification. Violations of these assumptions may lead to undercoverage or overly conservative intervals in practice. Quantile regression methods relax many of these assumptions by estimating conditional quantiles directly. In particular, quantile regression forests allow the interval width to vary flexibly across the covariate space, meaning that they can adapt naturally to heteroskedasticity, asymmetry, and nonlinear structure in the data. This local adaptivity allows the interval width to adjust to local variation in predictive uncertainty when the conditional distribution of the response depends on the covariates. However, these methods generally do not provide finite-sample validity guarantees, and their performance depends critically on accurate estimation of the conditional quantiles. Conformal prediction takes a fundamentally different approach by constructing prediction intervals using the empirical quantiles of nonconformity scores computed from the observed data. Under exchangeability, conformal methods provide exact finite-sample marginal validity independently of the underlying prediction model and without requiring distributional assumptions such as Gaussianity or homoskedasticity. At the same time, the conformal guarantee is marginal and does not ensure that the desired coverage level is achieved conditionally on the covariates. In particular, this may lead to inefficient or locally miscalibrated prediction intervals in heteroskedastic settings. Consequently, although conformal prediction guarantees validity, the efficiency and local adaptivity of the resulting intervals depend strongly on the underlying prediction model and the structure of the data-generating process.

Overall, classical methods achieve strong conditional guarantees under restrictive assumptions, quantile regression provides increased flexibility but weaker theoretical guarantees, while conformal prediction provides robust finite-sample validity but does not generally guarantee local adaptivity. The theoretical discussion above highlights several important practical questions. How sensitive are the different methods to model misspecification and heteroskedasticity? To what extent does increased flexibility improve interval efficiency while maintaining adequate coverage? And how pronounced are the differences between marginal and conditional validity in practice? These questions are investigated empirically through the simulation study and case study presented in the following chapters.

3 Empirical Validation

The theoretical results developed in Section 2 characterize the assumptions and validity guarantees underlying the considered prediction interval methods. In this chapter, these theoretical properties are investigated empirically through repeated simulation experiments under controlled data-generating mechanisms.

The purpose of the empirical study is to examine how the different prediction interval methods behave in finite samples under varying forms of model misspecification and distributional complexity. In particular, we study the trade-off between validity, efficiency, and local adaptivity across methods with different levels of model flexibility and distributional assumptions.

We assess how closely the resulting prediction intervals approximate the oracle interval, how robust the methods are to model misspecification and heteroskedasticity, and how their performance evolves as the sample size increases. To investigate these questions systematically, we consider four experimental settings designed to isolate different statistical challenges, including nonlinear regression structures, asymmetric noise, and heteroskedasticity.

3.1 Preliminaries

We first introduce the notation and evaluation criteria used throughout the empirical study. In particular, we distinguish between different types of quantiles used in the construction of prediction intervals and define the performance measures used to evaluate validity and efficiency across the simulation experiments.

Throughout the empirical study, we distinguish between the population quantile $q_\alpha(x)$, denoting the true α -quantile of $Y \mid X = x$, the estimated conditional quantile $\hat{q}_\alpha(x)$, obtained from a prediction model, and the empirical quantile $\hat{q}_\alpha$, computed from residuals or calibration scores.

For each simulation experiment, a training dataset is generated from the underlying data-generating process, the prediction intervals are constructed, and performance is evaluated on an independent test set $\{(X_i, Y_i)\}_{i=1}^{n_{\text{test}}}$.

Let

$$C(X_i) = [L(X_i), U(X_i)]$$

denote the prediction interval associated with the test observation (X_i, Y_i) .

The validity performance measure is the empirical coverage,

$$\hat{C} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \mathbb{1}\{Y_i \in C(X_i)\},$$

that is, the proportion of test observations contained in the prediction interval.

For a fixed realized training and calibration dataset $\mathcal{D}_n$, the prediction interval $C(\cdot)$ is fixed, and the corresponding training-conditional coverage is

$$p_n = \mathbb{P}(Y \in C(X) \mid \mathcal{D}_n).$$

Conditionally on $\mathcal{D}_n$, the indicators $\mathbf{1}\{Y_i \in C(X_i)\}$ are i.i.d. Bernoulli random variables with success probability p_n .

Hence,

$$\mathbb{E}[\hat{C} \mid \mathcal{D}_n] = p_n,$$

with conditional variance

$$\text{Var}(\hat{C} \mid \mathcal{D}_n) = \frac{p_n(1 - p_n)}{n_{\text{test}}}.$$

Thus, the empirical coverage satisfies

$$\hat{C} \rightarrow p_n \quad \text{as} \quad n_{\text{test}} \rightarrow \infty.$$

In other words, the empirical coverage observed in each simulation repetition provides a highly accurate approximation of the corresponding training-conditional coverage for large n_{test} (Vovk, 2012; Hjort et al., 2024a).

Each experiment is repeated independently M times using newly generated datasets. Averaging the empirical coverages across simulation repetitions yields the estimator

$$\bar{C} = \frac{1}{M} \sum_{j=1}^M \hat{C}^{(j)},$$

which approximates the marginal coverage

$$\mathbb{P}(Y \in C(X)).$$

Compared to the estimation of the training-conditional coverage within a single simulation run, estimation of the marginal coverage additionally depends on averaging over the randomness of the generated training datasets. Therefore, it typically requires a larger number of simulation repetitions to achieve numerical stability. This is also reflected in the theoretical results of Section 2.4.3, where the training-conditional coverage was shown to converge to $1 - \alpha$, implying that the variability arising within a fixed training dataset decreases as n grows.

For a single simulation repetition, the efficiency performance measure is the average interval length,

$$\hat{\mathcal{L}} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (U(X_i) - L(X_i)).$$

Across repeated simulation runs, the reported interval length is obtained through

$$\bar{\mathcal{L}} = \frac{1}{M} \sum_{j=1}^M \hat{\mathcal{L}}^{(j)}.$$

The relative interval length is defined as

$$\frac{\bar{\mathcal{L}}}{\bar{\mathcal{L}}_{\text{oracle}}},$$

where $\bar{\mathcal{L}}_{\text{oracle}}$ denotes the average oracle interval length across the simulation repetitions.

To jointly assess validity and efficiency, we additionally consider the Winkler score (Winkler, 1972). For a $(1 - \alpha)$ -prediction interval $C(X_i) = [L(X_i), U(X_i)]$, the Winkler score for observation i is defined as

$$W_i = (U(X_i) - L(X_i)) + \frac{2}{\alpha}(L(X_i) - Y_i)\mathbb{1}\{Y_i < L(X_i)\} + \frac{2}{\alpha}(Y_i - U(X_i))\mathbb{1}\{Y_i > U(X_i)\}.$$

The score rewards narrow prediction intervals while imposing a penalty whenever the observation falls outside the interval. Consequently, lower Winkler scores indicate better overall predictive performance.

For each simulation repetition, the average Winkler score is computed over the independent test set,

$$\hat{W} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} W_i,$$

and the reported score is obtained by averaging across the M simulation repetitions,

$$\bar{W} = \frac{1}{M} \sum_{j=1}^M \hat{W}^{(j)}.$$

3.2 Compared Methods

We compare several prediction interval methods representing the different theoretical frameworks introduced in Section 2. Specifically, we consider classical Gaussian prediction intervals from Section 2.2, quantile regression methods from Section 2.3, and conformal prediction methods from Section 2.4. In addition, we include a simple residual-quantile (RQ) baseline, which constructs prediction intervals from empirical residual quantiles of calibration residuals without applying conformal inference. Consequently, the method does not provide finite-sample validity guarantees and relies on the residual distribution being representative of future prediction errors. We also include the oracle interval, which serves as an idealized benchmark representing the optimal achievable interval efficiency.

The compared methods differ both in their validity guarantees and in their ability to adapt to the conditional distribution of $Y \mid X = x$. In particular, some methods rely on global residual quantiles, meaning that the same residual quantiles are used for all values of x , whereas others estimate conditional quantiles directly and therefore allow the interval width to vary across the covariate space. For the residual-based and conformal methods, we consider both linear regression (LM) and random forest (RF) regression estimators introduced in Section 2.1. For quantile regression methods, both linear quantile regression and quantile

regression forests are considered. This allows us to separately investigate the effect of the interval construction principle and the flexibility of the underlying prediction model. The classical Gaussian prediction interval introduced in Section 2.2 is considered only together with linear regression, since the interval construction relies on the analytical variance formula and Gaussian error assumptions underlying the linear model.

List of methods:

- **Oracle:** The oracle interval is defined as

$$C^*(x) = [q_{\alpha/2}(x), q_{1-\alpha/2}(x)],$$

where the conditional quantiles are computed from the true data-generating distribution.

- **Classical linear prediction interval (Gaussian linear):** The classical Gaussian prediction interval introduced in Section 2.2 is defined as

$$\hat{\mu}(X_{n+1}) \pm t_{n-p, 1-\alpha/2} \cdot \hat{\sigma} \sqrt{1 + X_{n+1}^\top (X^\top X)^{-1} X_{n+1}},$$

where $t_{n-p, 1-\alpha/2}$ denotes $(1 - \alpha/2)$ -quantile of the Student t -distribution with $n - p$ degrees of freedom. The interval achieves exact finite-sample validity under the Gaussian linear model assumptions, including linearity, homoskedasticity, and Gaussian errors.

- **Residual quantiles (RQ):** The residual quantile interval uses empirical quantiles of the residuals

$$R_i = Y_i - \hat{\mu}(X_i),$$

and constructs the interval using the empirical $\alpha/2$ and $1 - \alpha/2$ quantiles of the residual distribution $\{R_i\}$:

$$[\hat{\mu}(x) + \hat{q}_{\alpha/2}, \hat{\mu}(x) + \hat{q}_{1-\alpha/2}].$$

The method does not rely on Gaussianity and naturally accommodates asymmetric noise distributions.

- **Quantile regression (QR):** The quantile regression method introduced in Section 2.3 estimates conditional quantiles directly:

$$[\hat{q}_{\alpha/2}(x), \hat{q}_{1-\alpha/2}(x)].$$

In contrast to residual-based methods, the quantiles depend on x , allowing the interval to adapt to heteroskedasticity and asymmetry.

- **Split conformal prediction (CP):** The split conformal prediction interval introduced in Section 2.4 uses absolute residuals from a calibration set

$$R_i = |Y_i - \hat{\mu}(X_i)|,$$

and defines

$$C_{\text{CP}}(x) = [\hat{\mu}(x) - \hat{q}_{1-\alpha}, \hat{\mu}(x) + \hat{q}_{1-\alpha}].$$

Here, $\hat{q}_{1-\alpha}$ denotes the conformal quantile defined in Section 2.4. The method assumes exchangeability of the observations and provides finite-sample marginal validity.

3.3 Experimental Setup

We consider sample sizes

$$n \in \{20, 80, 320, 1280\},$$

increasing geometrically by a factor of four. The target coverage level is fixed at $1 - \alpha = 0.9$. Each dataset is split evenly into training and calibration sets. Performance is evaluated on an independent test set of size $n_{\text{test}} = 100,000$, and each experiment is repeated $M = 200$ times.

We consider four experimental settings designed to isolate different forms of model misspecification and distributional complexity. Throughout the simulation study, we set

$$\beta = 2, \quad a = 1, \quad \sigma = 1.$$

Unless otherwise stated, these parameter values are used in all experiments.

Experiment 1: Linear model with Gaussian noise

The data are generated according to

$$Y = \beta X + a + \varepsilon,$$

where

$$X \sim \text{Unif}([-1, 1]), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad X \perp\!\!\!\perp \varepsilon.$$

This setting represents the benchmark scenario in which all assumptions of the classical Gaussian linear model are satisfied.

Experiment 2: Linear model with asymmetric noise

To evaluate robustness to skewed error distributions, we replace the Gaussian noise with a shifted exponential distribution:

$$\varepsilon \sim \text{Exp}(1/\sigma) - \sigma,$$

The centering ensures that $\mathbb{E}[\varepsilon] = 0$ and $\text{Var}(\varepsilon) = \sigma^2$, allowing the effect of asymmetry to be studied independently of changes in the mean and variance. However, the distribution of ε is asymmetric in this case. Its quantiles are given by

$$q_\alpha = -\sigma \log(1 - \alpha) - \sigma,$$

which are not symmetric around zero, i.e. $q_{\alpha/2} \neq -q_{1-\alpha/2}$. Consequently, the oracle prediction interval becomes asymmetric, with a larger upper than lower tail adjustment.

Experiment 3: Nonlinear model with Gaussian noise

To study model misspecification, the conditional mean is replaced by the nonlinear function

$$\mu(x) = \sin(2\pi x),$$

while the noise remains Gaussian:

$$\varepsilon \sim \mathcal{N}(0, \sigma^2).$$

This setting favors more flexible estimators such as random forests.

Experiment 4: Linear model with heteroskedastic noise

Finally, we consider a heteroskedastic regression model of the form

$$Y = \beta X + a + \sigma(X)\varepsilon, \quad \sigma(x) = 1 + x,$$

where

$$\varepsilon \sim \mathcal{N}(0, 1).$$

Conditional on $X = x$, we therefore have

$$Y \mid X = x \sim \mathcal{N}(\mu(x), \sigma^2(x)),$$

such that

$$\text{Var}(Y \mid X = x) = \sigma^2(x) = (1 + x)^2.$$

Hence, the variability of the response increases with x , implying that the oracle prediction interval becomes progressively wider for larger values of x . Therefore, this setting tests whether the prediction interval methods can adapt locally to varying noise levels.

Random Forest Tuning

The random forest-based methods are implemented using the **ranger** package (Wright and Ziegler, 2017), which is used both for random forest regression and quantile regression forests.

The primary tuning parameter is the minimum node size (**min.node.size**), defined as the minimum number of observations required in a terminal node. This parameter controls the complexity of the fitted trees and thereby the bias-variance trade-off. Smaller node sizes allow the forest to capture local structure more flexibly, while larger node sizes produce smoother and more stable predictions (Hastie et al., 2009). The optimal choice of **min.node.size** depends on both the sample size and the underlying data-generating process. Consequently, the parameter is tuned separately for each method, simulation setting, and sample size using different grids of candidate values. Particular care was taken when tuning quantile regression forests, whose empirical coverage was found to be especially sensitive to the choice of **min.node.size**.

The remaining random forest parameters are kept fixed throughout the simulation study. The number of trees is set to (**B**=500), which was found to provide stable predictions while keeping the computational burden manageable. The **mtry** parameter, controlling the number of candidate variables considered at each split, and the subsampling fraction, controlling the proportion of observations used to grow each tree, are both kept at their default **ranger** settings to limit the tuning complexity and focus on the parameter with the largest observed impact on interval performance, namely **min.node.size**.

3.4 Results

Figure 1 provides an overall summary of the trade-off between empirical coverage and interval efficiency across all experimental settings for both small and large sample sizes. Methods located closer to the upper-left corner simultaneously achieve high coverage and short intervals and are therefore preferable. The individual experimental results are analyzed in greater detail in Section 3.4.1.

As described in Section 3.3, all simulations are conducted for four different sample sizes,

$$n \in \{20, 80, 320, 1280\}.$$

However, to improve readability, Figure 1 only displays the smallest and largest sample sizes. The intermediate sample sizes exhibit similar monotone convergence behaviour as n increases, provided that the underlying models are appropriately tuned. As a result, the figure primarily illustrates the finite-sample behaviour for small samples together with the approximate large-sample behaviour of the different methods.

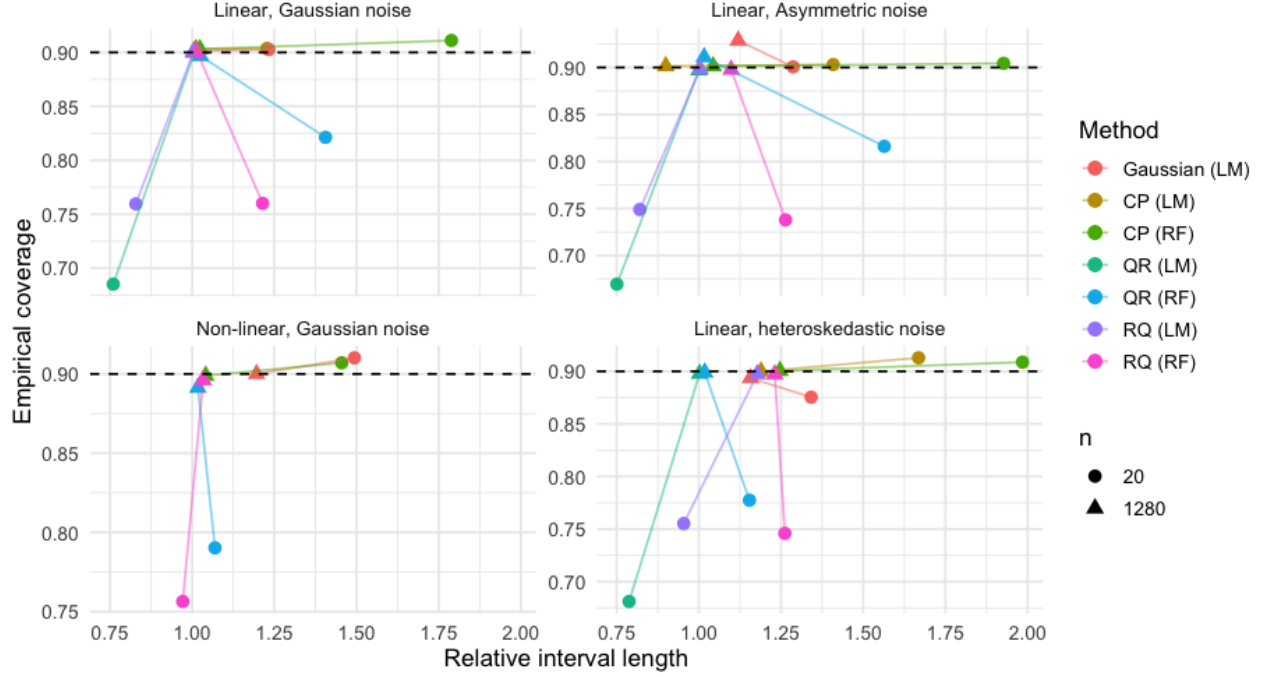


Figure 1: Empirical coverage versus relative interval length across the four experimental settings. The dashed horizontal line indicates the nominal coverage level of $1 - \alpha = 0.9$. Interval lengths are normalized relative to the oracle interval, such that values closer to one indicate intervals closer to the optimal benchmark.

Several overall patterns emerge from Figure 1. First, conformal prediction methods achieve coverage levels closest to the nominal target across all settings, confirming the finite-sample robustness predicted by the theoretical results. However, this robustness often comes at the cost of wider intervals, particularly in the asymmetric and heteroskedastic settings.

Second, Gaussian prediction intervals perform very well when the model assumptions are correctly specified, producing intervals close to the oracle benchmark in the linear Gaussian setting. However, the performance deteriorates once the assumptions are violated. In particular, heteroskedastic settings lead to undercoverage in finite samples and wider intervals, illustrating the sensitivity of parametric Gaussian intervals to misspecification.

The results furthermore demonstrate the importance of flexible regression estimators in non-linear settings. In Experiment 3, all methods are based on random forest estimators, since linear models are inherently misspecified in this setting. The results show that even when flexible machine learning models are able to capture the nonlinear structure of the conditional mean and quantiles, several methods still exhibit undercoverage for small sample sizes. Additionally, none of the methods attain the oracle interval length, even at the largest sample size considered.

The results also highlight the importance of adaptivity to the conditional distribution. Methods based on conditional quantile estimation generally perform better in asymmetric and heteroskedastic settings, where interval width varies across the covariate space. Nevertheless, these methods may still exhibit noticeable finite-sample deviations from the nominal coverage level.

Finally, increasing the sample size improves both coverage stability and interval efficiency across all methods. In particular, the asymptotic properties of quantile regression methods become increasingly visible for larger sample sizes.

To complement the separate analysis of empirical coverage and interval efficiency, Figure 2 reports the average Winkler score across all experimental settings and sample sizes. Unlike Figure 1, which visualizes the trade-off between coverage and relative interval length directly, the Winkler score combines both aspects into a single summary measure of prediction interval quality. Lower values therefore indicate methods that simultaneously achieve narrow intervals and coverage levels close to the nominal target.

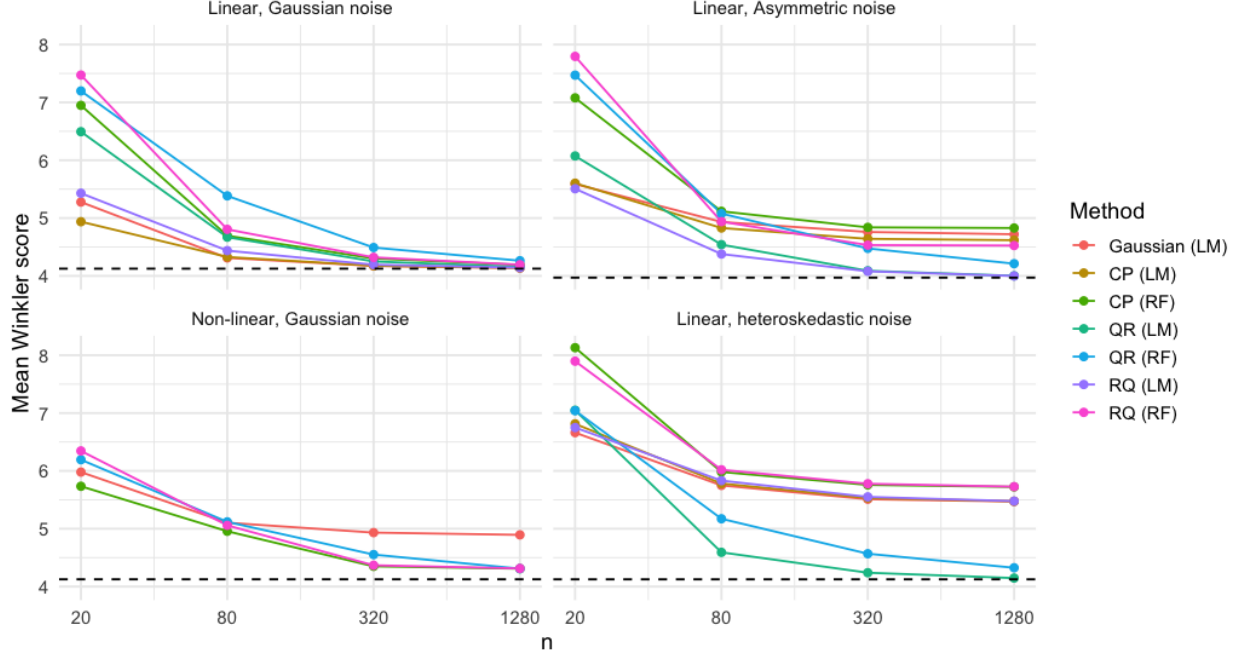


Figure 2: Mean Winkler scores across the four experimental settings as a function of the sample size, n . The dashed horizontal line indicates the corresponding Oracle benchmark. Lower Winkler scores indicate better overall performance.

Several general patterns emerge from Figure 2. First, all methods improve as the sample size increases, with the largest reduction in the Winkler score occurring between $n = 20$ and $n = 80$. Beyond this point, the improvements become more gradual, indicating convergence toward their asymptotic behaviour.

In the linear Gaussian setting, most methods approach the Oracle benchmark relatively closely for large sample sizes, reflecting that the underlying model assumptions are correctly specified. In contrast, the asymmetric and heteroskedastic settings remain considerably more challenging, where several methods continue to exhibit noticeably larger Winkler scores even for large n . This reflects the increased difficulty of simultaneously maintaining valid coverage while avoiding excessively wide intervals under model misspecification and varying conditional variance. Methods based on conditional quantile estimation generally perform better in asymmetric and heteroskedastic settings, where the interval width varies across the covariate space, while the conformal prediction methods achieve some of the lowest Winkler scores in finite samples when the model is homoskedastic.

Overall, the Winkler score results are consistent with the conclusions obtained from the coverage-relative length analysis. Conformal prediction methods remain among the most robust approaches in terms of validity, although this robustness is often associated with wider intervals. Parametric Gaussian intervals remain highly efficient under correct model assumptions but deteriorate under misspecification, while quantile regression and machine learning-based methods provide greater adaptivity in more complex data-generating settings.

Taken together, the simulation study demonstrates that no single method uniformly dominates across all scenarios. Classical Gaussian linear intervals are highly efficient under correct assumptions but fragile under misspecification, while conformal prediction provides robust validity guarantees at the expense of wider intervals. Methods based on conditional quantile estimation offer greater adaptivity to complex data-generating mechanisms, but their practical performance depends strongly on model quality and sample size.

While the overall comparison provides a broad picture of the trade-offs between validity and efficiency, several important patterns deserve a more detailed discussion. In the following sections, we therefore examine each experimental setting separately and analyze the most notable behaviors of the different prediction interval methods under varying forms of model misspecification and distributional complexity.

3.4.1 Detailed Analysis

Experiment 1: Linear model with Gaussian noise

The first experiment serves as a baseline setting in which all model assumptions are satisfied. Since the data are generated from a linear model with homoskedastic Gaussian noise, the Gaussian prediction interval coincides with the oracle interval asymptotically.

As expected, all methods achieve empirical marginal coverage close to the nominal level for sufficiently large sample sizes and produce intervals with lengths close to the oracle benchmark. The Gaussian interval performs particularly well, reflecting that simple parametric methods are highly efficient when the underlying assumptions are correct.

The conformal prediction methods achieve marginal coverage closest to the nominal target across all sample sizes, consistent with the finite-sample validity guarantee. However, this comes at the cost of slightly wider intervals. In contrast, QR and RQ exhibit mild under-coverage for small sample sizes, and their empirical marginal coverage improves rapidly as n increases and the estimated quantiles become more stable.

The random forest-based methods are generally less efficient than the linear methods for small sample sizes, as the additional flexibility introduces higher variance in a setting where the true model is linear. As the sample size increases, these differences gradually disappear.

Figure 3 illustrates representative prediction intervals from a single realization of Experiment 1.

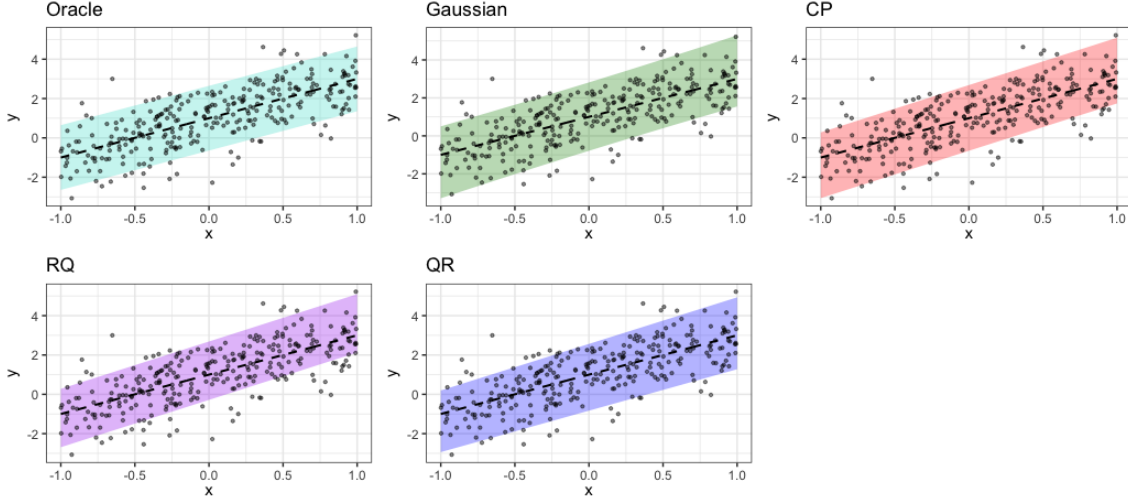


Figure 3: Prediction intervals based on a single realization of Experiment 1 with $n = 80$, using an independent visualization sample of size $n_{\text{test}} = 300$. The dashed line represents the true regression function. Only methods based on the linear prediction model are shown in this figure.

Overall, the experiment illustrates that when the model assumptions are correctly specified, simple parametric methods are both valid and highly efficient on average over the data-generating distribution, leaving limited advantage to more flexible approaches.

Experiment 2: Linear model with asymmetric noise

We next consider the setting with asymmetric noise generated from a shifted exponential distribution. In this case, the conditional distribution of $Y | X = x$ is right-skewed, implying that the oracle prediction interval is asymmetric with a longer upper tail. Figure 4 illustrates this behavior for a representative simulated dataset.

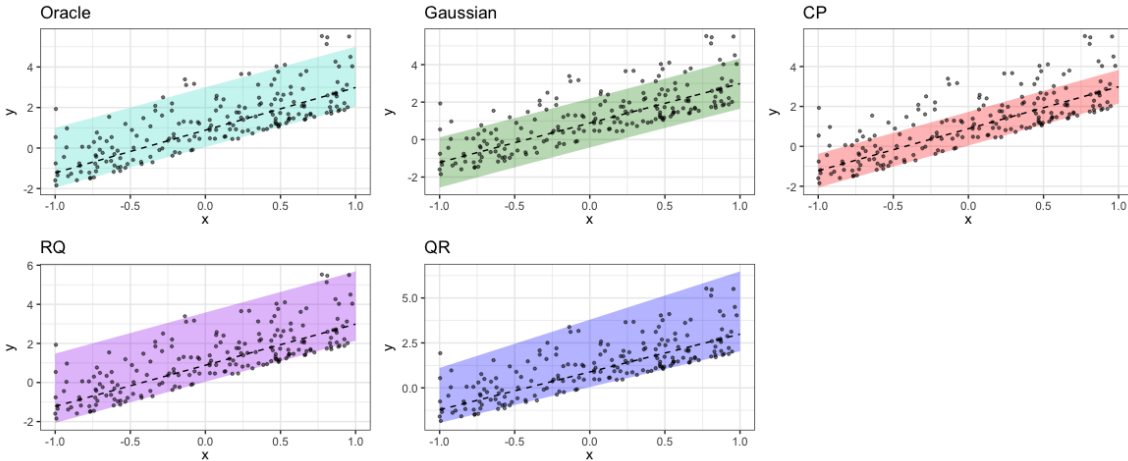


Figure 4: Prediction intervals based on a single realization of the experiment with asymmetric noise with $n = 80$, using an independent visualization set of size $n_{\text{test}} = 300$. The dashed curve represents the true regression function. Only methods based on the linear prediction model are shown in this figure.

Figure 4 shows explicitly that both Gaussian prediction intervals and conformal prediction produce symmetric intervals around the estimated conditional mean. As a result, these methods fail to adapt fully to the skewness of the conditional distribution. Although the intervals produced by QR and RQ appear visually reasonable for a single simulated dataset, the empirical coverage results reveal undercoverage when averaged across repeated simulations for small sample sizes, as shown in Figure 1.

The conformal prediction methods achieve empirical marginal coverage closest to the nominal target across all sample sizes, demonstrating the robustness of the conformalization even under distributional misspecification. However, this robustness comes at the cost of wider intervals, particularly for small n , where the conformal quantile must compensate for both skewness and estimation uncertainty.

Overall, the experiment highlights the importance of modelling the conditional distribution beyond the mean alone. In particular, symmetric residual-based methods may become inefficient or poorly calibrated under asymmetric noise, whereas quantile regression approaches are able to adapt to the skewed distributional structure.

Experiment 3: Nonlinear model with Gaussian noise

We next consider a nonlinear regression function, where the linear model is misspecified by construction. In this setting, the main challenge is no longer the noise distribution itself, but rather the inability of simple parametric models to capture the underlying conditional mean structure. Figure 5 illustrates the resulting prediction intervals for the different methods.

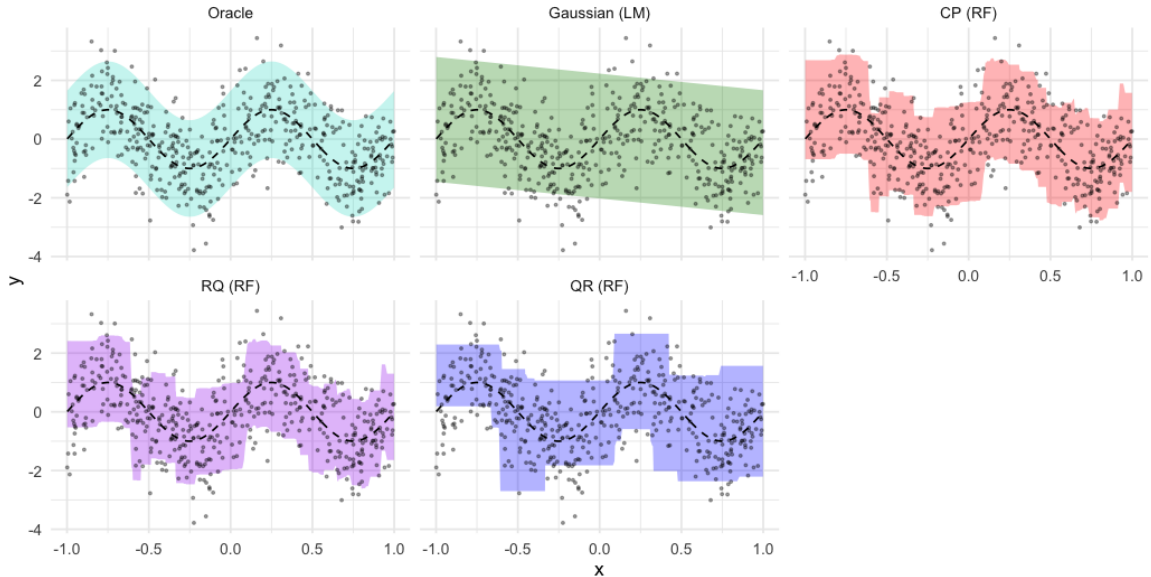


Figure 5: Prediction intervals based on a single realization in the nonlinear setting with $n = 80$, using an independent visualization set of size $n_{\text{test}} = 500$. The dashed curve represents the true regression function $f(x) = \sin(2\pi x)$. Only methods based on the random forest prediction model are shown in this figure.

As expected, the Gaussian linear model performs poorly in this setting. Since the conditional mean is nonlinear, the fitted linear model introduces systematic bias, which propagates directly to the prediction intervals. Although the Gaussian intervals remain approximately constant in width, they fail to follow the nonlinear structure of the data, resulting in noticeably larger widths than the oracle intervals.

The nonlinear setting furthermore highlights the importance of flexible regression estimators. Both conformal prediction and quantile regression based on random forests are able to adapt to the nonlinear conditional mean and produce intervals that closely follow the true regression function. In particular, the conformal intervals maintain empirical marginal coverage close to the nominal level across all sample sizes, illustrating the robustness of conformal prediction under model misspecification.

However, the results also demonstrate that accurate prediction of the conditional mean alone does not guarantee efficient uncertainty quantification. While the random forest-based methods successfully capture the nonlinear structure, the estimated intervals still differ in both coverage and relative length for small sample sizes. In particular, QR and RQ exhibit noticeable undercoverage when n is small, despite visually adapting well to the nonlinear trend. This illustrates the finite-sample variability of estimated conditional quantiles and the difficulty of accurately estimating tail behavior in limited samples.

Figure 6 further illustrates the local behavior of the different methods. The left panel shows the interval length as a function of x , while the right panel shows the corresponding conditional coverage,

$$\mathbb{P}(Y \in C(X) \mid X = x).$$

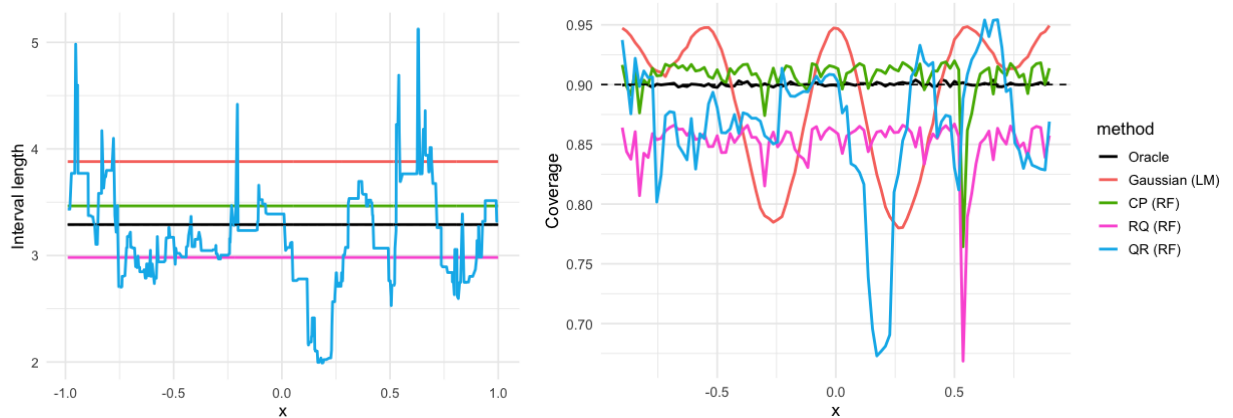


Figure 6: Local interval diagnostics in the nonlinear setting based on a single realization with $n = 1280$. The left panel shows interval lengths as a function of x , while the right panel shows the corresponding conditional coverage estimated at fixed covariate values.

The oracle interval has constant width due to the homoskedastic Gaussian noise, whereas the random forest quantile regression intervals exhibit more local variability for small sample sizes. In particular, the quantile-regression-based methods display unstable conditional coverage across the covariate space, reflecting the difficulty of accurately estimating conditional tail quantiles from limited data.

The Gaussian intervals, by contrast, appear relatively stable in width, but their conditional coverage varies systematically across x . This occurs because the intervals are centered around a misspecified linear regression function rather than the true nonlinear conditional mean. Consequently, the Gaussian method may still achieve valid marginal coverage on average, while failing locally in regions where the model bias is largest.

Overall, the nonlinear experiment demonstrates that flexible machine learning estimators are essential when the regression function is misspecified. At the same time, the results show that adaptivity alone is insufficient: valid uncertainty quantification additionally requires careful control of finite-sample variability.

Experiment 4: Linear model with heteroskedastic noise

In the heteroskedastic setting, the uncertainty now varies across the covariate space. Methods based on globally estimated residual quantiles, such as Gaussian prediction intervals, residual quantiles, and conformal prediction, are unable to adapt their interval widths locally. As a result, these methods produce intervals with constant width even though the oracle interval expands with x .

This behavior leads to systematic conditional miscoverage. For smaller values of x , the constant-width intervals tend to overcover because the local noise level is relatively small. Conversely, for larger values of x , the intervals become too narrow relative to the true conditional variance and therefore undercover. Although conformal prediction still maintains approximate marginal validity on average, the local coverage behavior varies considerably across the covariate space.

By contrast, quantile regression based on random forests is able to adapt more directly to the changing conditional distribution, resulting in interval lengths that increase with x , more closely resembling the oracle behavior. However, this flexibility comes at the cost of increased variability in the estimated conditional quantiles, especially for smaller sample sizes. Consequently, the local interval lengths and conditional coverage fluctuate considerably across the covariate space. Figure 7 illustrates these effects directly.

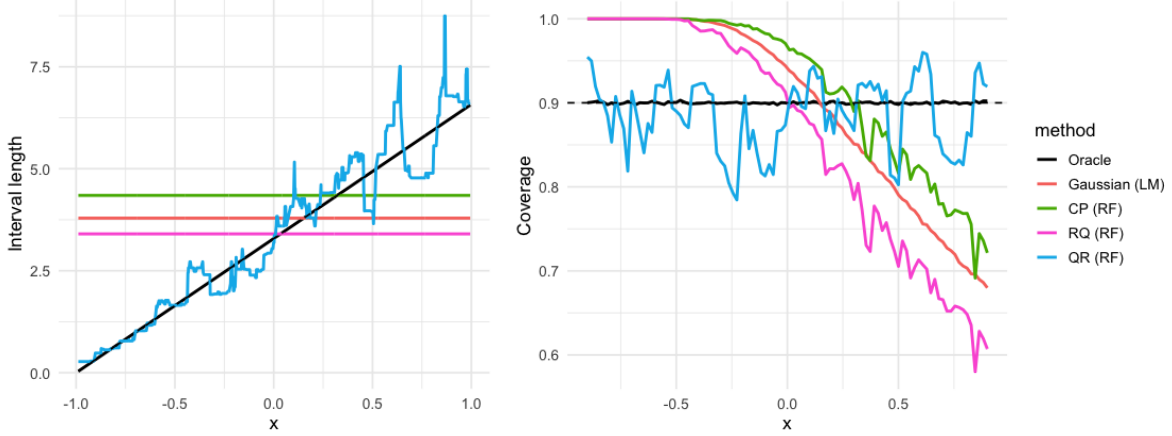


Figure 7: Local interval diagnostics in the heteroskedastic setting based on a single realization with $n = 1280$. The left panel shows interval lengths as a function of x , while the right panel shows the corresponding conditional coverage estimated at fixed covariate values.

Figure 7 shows that the oracle interval length increases linearly with x , reflecting the increasing conditional variance. In contrast, Gaussian prediction intervals, conformal prediction, and residual-quantile methods remain nearly constant in width and, as a result, fail to adapt locally to the heteroskedastic noise structure. This directly translates into the systematic decline in conditional coverage observed for larger values of x . The quantile-regression-based intervals exhibit the opposite behavior. Their interval lengths adapt more closely to the oracle benchmark, but the estimated conditional quantiles become considerably more unstable in finite samples. As a result, the conditional coverage fluctuates across the covariate space despite the improved adaptivity. Figure 8 shows representative prediction intervals from a single realization of Experiment 4.

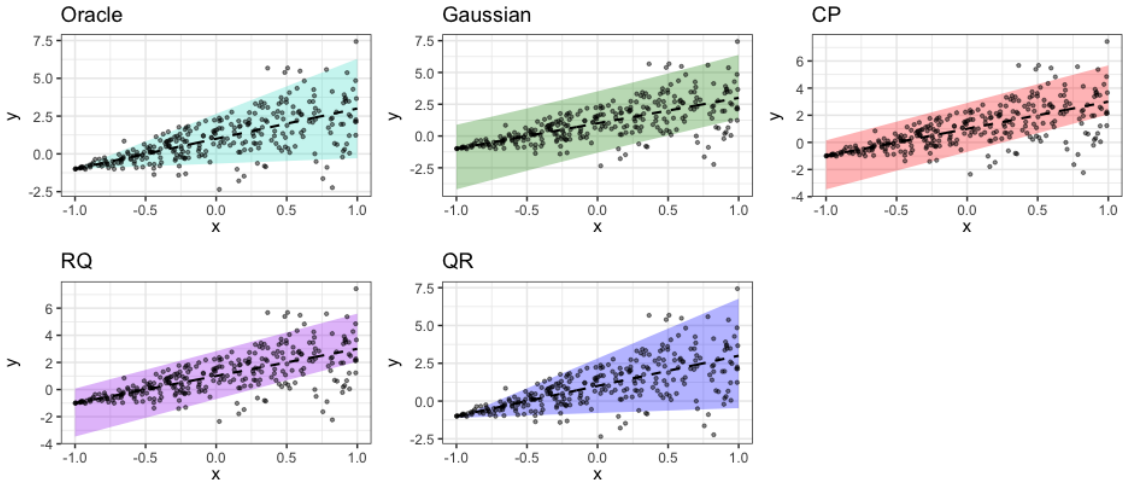


Figure 8: Prediction intervals based on a single realization of Experiment 4 with $n = 80$, using an independent visualization sample of size $n_{\text{test}} = 300$. The dashed line represents the true regression function. Only methods based on the linear prediction model are shown in this figure.

Overall, the heteroskedastic experiment illustrates a fundamental limitation of globally calibrated prediction intervals. Even methods with marginal validity guarantees may fail locally when the conditional distribution varies with the covariates. At the same time, the results demonstrate that locally adaptive methods require sufficiently accurate estimation of conditional quantiles in order to achieve stable uncertainty quantification.

3.5 Discussion

Taken together, the four experiments illustrate that no single method uniformly dominates across all settings. Instead, the results reveal a fundamental trade-off between finite-sample validity, adaptivity, and interval efficiency. For consistency across methods, the same train/-calibration/test splitting procedure was used for all approaches. Since Gaussian prediction intervals and quantile regression intervals do not require calibration, the calibration subset was not used during model fitting for these methods. In practice, these methods could therefore potentially achieve better predictive performance if the calibration observations were incorporated directly into the model estimation procedure.

Methods based on global residual quantiles, such as Gaussian prediction intervals, residual quantiles, and conformal prediction, generally provide stable marginal validity across a wide range of settings. However, the experiments also show that these methods struggle when the conditional distribution varies across the covariate space. In particular, the heteroskedastic setting demonstrates that globally calibrated intervals may achieve average validity while simultaneously exhibiting severe local miscoverage. Conversely, methods that explicitly model conditional quantiles are generally more adaptive. In particular, quantile regression forests are able to accommodate nonlinear structure and heteroskedasticity, often yielding interval lengths closer to the oracle benchmark in complex settings. However, the simulation results also demonstrate that adaptivity alone does not guarantee valid uncertainty quantification. In several experiments, the quantile regression methods exhibit noticeable finite-sample undercoverage despite accurately tracking the conditional structure of the data.

Therefore, the experiments highlight a central limitation of the methods considered so far. Globally calibrated methods provide validity but insufficient adaptivity, whereas locally adaptive methods improve efficiency but lack finite-sample validity guarantees. This naturally motivates methods that combine both principles simultaneously.

3.6 Extension: Conformalized Quantile Regression

To investigate whether conformal prediction can improve the finite-sample validity of adaptive quantile intervals, the previous simulation experiments are repeated with the addition of conformalized quantile regression (CQR) (Romano et al., 2019), which combines quantile regression with conformal prediction.

Let $(X_1, Y_1), \dots, (X_n, Y_n), (X_{n+1}, Y_{n+1})$ be exchangeable observations. As in split conformal prediction, the data are divided into two disjoint subsets, I_1 and I_2 , of equal size, i.e. $|I_1| =$

$|I_2| = n/2$. Using the data in I_1 , estimate the conditional quantile functions,

$$\hat{q}_{\alpha/2}(x), \quad \hat{q}_{1-\alpha/2}(x).$$

In the empirical experiments, the conditional quantiles are estimated using quantile regression forests. On the calibration set I_2 , define the nonconformity score

$$E_i = \max\left\{\hat{q}_{\alpha/2}(X_i) - Y_i, Y_i - \hat{q}_{1-\alpha/2}(X_i)\right\}, \quad i \in I_2.$$

The conformal correction is then obtained through the empirical conformal quantile

$$\hat{q}_{1-\alpha} = E_{(k^*)}, \quad k^* = \lceil (n/2 + 1)(1 - \alpha) \rceil.$$

The resulting conformalized quantile regression interval is

$$C_{\text{CQR}}(x) = \left[\hat{q}_{\alpha/2}(x) - \hat{q}_{1-\alpha}, \hat{q}_{1-\alpha/2}(x) + \hat{q}_{1-\alpha}\right].$$

By exchangeability,

$$\mathbb{P}(Y_{n+1} \in C_{\text{CQR}}(X_{n+1})) \geq 1 - \alpha.$$

Thus, CQR preserves the finite-sample marginal validity of split conformal prediction while simultaneously allowing the interval width to adapt locally through conditional quantile estimation (Romano et al., 2019; Sesia and Candès, 2019).

Figure 9 summarizes the empirical coverage and relative interval lengths obtained after adding conformalized quantile regression to the simulation experiments. To facilitate comparison, the heatmap focuses on the three random-forest-based methods: quantile regression (QR), split conformal prediction (CP), and conformalized quantile regression (CQR).



Figure 9: Empirical coverage (left) and relative interval lengths (right) for CP (RF), QR (RF), and CQR (RF) across the four experimental settings and sample sizes. Coverage values closer to the nominal target level of $1 - \alpha = 0.9$ are shown in green, while undercoverage is indicated by warmer colors. Relative interval lengths are normalized relative to the oracle interval length, such that values closer to 1 indicate intervals closer to the optimal benchmark.

Overall, the results show that conformalization improves the finite-sample validity of the quantile regression intervals. In particular, the pronounced undercoverage previously observed for quantile regression forest at small sample sizes is largely eliminated after conformal calibration. Across all experimental settings, CQR achieves empirical coverages close to the nominal level even for moderate sample sizes.

At the same time, the resulting interval lengths remain closer to the oracle benchmark than those obtained using standard split conformal prediction. Since the interval width in CQR is driven by estimated conditional quantiles rather than a single global residual quantile, the intervals retain local adaptivity and continue to adjust to heterogeneous uncertainty across the covariate space. This effect is particularly pronounced in the heteroskedastic settings, where CQR produces shorter intervals than ordinary conformal prediction while still maintaining approximately valid coverage.

Nevertheless, slight undercoverage is still occasionally observed for small sample sizes. This behavior is likely inherited from the underlying quantile regression forest estimator. Since the conformal correction is constructed on top of the estimated lower and upper conditional quantiles, inaccurate or systematically overly narrow baseline quantile estimates may still affect finite-sample performance. In particular, if the estimated quantiles fail to capture the true local variability of the response, the conformal adjustment may not be sufficient to recover the nominal coverage level in finite samples fully.

Overall, CQR may be interpreted as a compromise between quantile regression and conformal prediction. Compared to ordinary quantile regression, conformal calibration improves finite-sample validity. Compared to standard split conformal prediction, however, the intervals remain locally adaptive through the estimated conditional quantiles and therefore retain considerably better efficiency in heteroskedastic settings. This explains why the empirical performance of CQR lies between quantile regression forests and conformal prediction in the simulations.

From a theoretical perspective, CQR modifies the quantile regression interval through a conformal prediction step, replacing the asymptotic validity properties of quantile regression with finite-sample marginal validity guarantees. While this generally improves finite-sample coverage, the resulting guarantees remain marginal and do not imply accurate conditional coverage. Detailed numerical results for all simulation experiments are reported in Appendix A.1.

4 Case study: Prediction Intervals for Housing Prices

This section presents an empirical case study based on the `CphHousingPrices` dataset, which contains transactions of single-family houses in the Capital Region of Denmark from 2000 to 2025. The dataset was kindly provided by the Danish Knowledge Centre for Housing Economics (*Boligøkonomisk Videncenter*) under an agreement regarding data access and usage. The goal is to construct and compare prediction intervals for housing prices using the methods developed throughout this thesis.

Accurate prediction intervals are particularly important in housing markets, where uncertainty and heterogeneity make point predictions alone insufficient for reliable valuation and risk assessment (Hjort et al., 2024b). Housing prices are influenced by a complex combination of structural, geographic, and temporal factors, and the associated prediction uncertainty may vary considerably across properties. Consequently, the construction of valid, efficient, and locally adaptive prediction intervals becomes especially relevant in this setting (Hjort et al., 2024a).

Specifically, we consider classical Gaussian prediction intervals based on a linear regression model from Section 2.2, split conformal prediction intervals from Section 2.4 based on random forest regression, and quantile regression intervals from Section 2.3 constructed using quantile regression forests. For clarity, the standard split conformal prediction method considered throughout the thesis will be referred to as global conformal prediction, distinguishing it from the localized Mondrian conformal prediction approach introduced in Section 2.4.3. The methods are compared in terms of empirical coverage and interval efficiency in order to assess how the theoretical trade-offs identified in the simulation study manifest in a real-world prediction problem.

In particular, the case study investigates whether the different methods achieve the desired nominal coverage level of 90%. At the same time, we examine how the interval lengths vary across methods and whether locally adaptive methods produce more informative prediction intervals in the presence of heterogeneous housing price uncertainty.

4.1 Data Description

The data consists of 227,515 observations with cash purchase price (DKK) as the response variable and 20 explanatory variables. Summary statistics for the variables included in the analysis are reported in Table 1.

Variable	Type	Mean	Median	Min	Max
Sale price (DKK)	Numerical	2,899,358	2,841,947	120,572	80,197,267
Residential area (m^2)	Numerical	140.3	134	25	400
Number of rooms	Numerical	4.8	5	0	20
Building age	Numerical	51.9	49	-21	275
Public valuation area (m^2)	Numerical	807.3	717	0	99,698
Number of bathrooms	Numerical	1.4	1	0	6
Number of toilets	Numerical	1.8	2	1	6
Number of floors	Numerical	1.1	1	1	4
Basement area (m^2)	Numerical	23.2	0	0	481
Distance to coast (m)	Numerical	5,231	4,461.8	12.9	15,536.1
Year of renovation	Numerical	645.4	0	0	2024
Year of sale	Numerical	2013	2013	2000	2025
Usage code	Categorical	122.9	120	110	190
Heating installation code	Categorical	2.2	2	1	99
Outer wall material code	Categorical	2.5	1	1	90
Roof material code	Categorical	5.1	4	1	90
Condemned dwelling	Binary	–	–	F	T
Toilet condition code	Binary	–	–	F	T
Bathroom condition code	Categorical	–	–	C	V
Region	Categorical	–	–	–	–
Municipality	Categorical	–	–	–	–

Table 1: Summary statistics for the variables used in the housing price case study. Numerical administrative codes are treated as categorical variables. In the table, F is an abbreviation for False, and T is an abbreviation for True.

4.1.1 Descriptive Analysis

The dataset displays variation in both housing prices and property characteristics. In particular, the raw housing prices exhibit right-skewness, motivating the logarithmic transformation shown in Figure 10. Furthermore, Figure 10 illustrates temporal and geographic variation in the housing market.

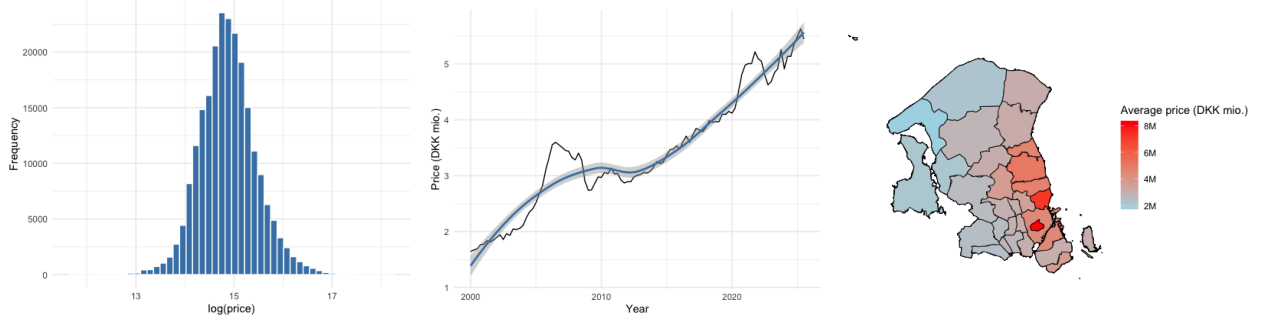


Figure 10: Descriptive characteristics of the housing dataset. **Left:** Histogram of log-transformed housing prices. **Middle:** Average housing prices over time, together with a smoothed trend curve. **Right:** Average housing prices across municipalities in the Capital Region of Denmark.

Figure 10 illustrates that average housing prices have increased considerably over time, while large differences are observed across municipalities within the Capital Region of Denmark. These characteristics make the dataset particularly suitable for studying prediction intervals. The combination of skewed prices, nonlinear relationships between covariates and prices, and changing variability across both time and location suggests that the assumptions of a simple homoskedastic linear model are unlikely to be satisfied. Consequently, the dataset provides a realistic setting for comparing classical Gaussian prediction intervals, adaptive quantile regression, and distribution-free conformal prediction approaches.

4.2 Methodology

Let

$$\{(X_i, Y_i)\}_{i=1}^n$$

denote the observed housing transactions, where $X_i \in \mathbb{R}^p$ contains covariates describing property i , and $Y_i \in \mathbb{R}_+$ denotes the corresponding sale price in DKK. Due to the strong right-skewness of housing prices, we model

$$\tilde{Y}_i = \log(Y_i).$$

As a consequence, all prediction intervals are constructed on the logarithmic scale and subsequently transformed back to the original price scale.

To preserve the temporal distribution of observations across the training, calibration, and test sets, the data splitting is performed independently within each year rather than chronologically across years. For each year t , let

$$I_t = \{i : \text{year}_i = t\}$$

denote the corresponding subset of observations. Within each year, the data is randomly split into three disjoint subsets,

$$I_t = I_t^{\text{train}} \cup I_t^{\text{cal}} \cup I_t^{\text{test}},$$

with approximate proportions

$$60\% \text{ training}, \quad 20\% \text{ calibration}, \quad 20\% \text{ testing}.$$

The final datasets are then obtained by combining the corresponding subsets across years:

$$I^{\text{train}} = \bigcup_t I_t^{\text{train}}, \quad I^{\text{cal}} = \bigcup_t I_t^{\text{cal}}, \quad I^{\text{test}} = \bigcup_t I_t^{\text{test}}.$$

This stratified splitting procedure reduces the influence of large temporal differences in housing market conditions while ensuring that all years remain represented in each subset.

Three prediction interval methods are considered. First, classical Gaussian prediction intervals are constructed using a linear regression model fitted on the logarithmic price scale. Second, split conformal prediction intervals are constructed using random forest regression as the underlying mean estimator. Third, quantile regression intervals are constructed using quantile regression forests, estimating the lower and upper conditional quantiles directly.

The random forest-based methods are implemented using the **ranger** package (Wright and Ziegler, 2017) with default parameter settings and $B = 500$ trees.

For the classical Gaussian prediction intervals, the interval on the logarithmic scale is constructed using the standard linear regression prediction interval as defined in Section 2.2,

$$\left[\hat{\mu}(x) - t_\alpha \cdot \hat{\sigma} \sqrt{1 + x^\top (X^\top X)^{-1} x}, \hat{\mu}(x) + t_\alpha \cdot \hat{\sigma} \sqrt{1 + x^\top (X^\top X)^{-1} x} \right],$$

where $t_\alpha = t_{n-p, 1-\alpha/2}$ denotes the $(1 - \alpha/2)$ -quantile of the Student t -distribution with $n - p$ degrees of freedom. Transforming back to the original scale yields

$$C_{\text{Gaus}}(x) = \left[\exp\left(\hat{\mu}(x) - t_\alpha \cdot \hat{\sigma} \sqrt{1 + x^\top (X^\top X)^{-1} x}\right), \exp\left(\hat{\mu}(x) + t_\alpha \cdot \hat{\sigma} \sqrt{1 + x^\top (X^\top X)^{-1} x}\right) \right].$$

For split conformal prediction, the interval on the logarithmic scale is given by

$$[\hat{\mu}(x) - \hat{q}_{1-\alpha}, \hat{\mu}(x) + \hat{q}_{1-\alpha}],$$

where $\hat{q}_{1-\alpha}$ denotes the conformal quantile of the absolute calibration residuals as defined in Section 2.4. Transforming back to the original scale yields

$$C_{\text{CP}}(x) = [\exp(\hat{\mu}(x) - \hat{q}_{1-\alpha}), \exp(\hat{\mu}(x) + \hat{q}_{1-\alpha})].$$

Finally, the quantile regression forest intervals introduced in Section 2.3 are constructed directly from the estimated conditional quantiles on the logarithmic scale,

$$[\hat{q}_{\alpha/2}(x), \hat{q}_{1-\alpha/2}(x)],$$

yielding the interval

$$C_{\text{QRF}}(x) = [\exp(\hat{q}_{\alpha/2}(x)), \exp(\hat{q}_{1-\alpha/2}(x))]$$

on the original price scale.

For comparability across methods, the Gaussian and quantile regression models are fitted using only the training set I^{train} , even though these approaches do not require a separate calibration dataset.

Unlike the controlled simulation experiments, the housing dataset does not fully satisfy the theoretical assumptions underlying the prediction interval methods. In particular, temporal market changes and potential distributional shifts over time imply that exact exchangeability cannot be guaranteed. Similarly, the assumptions underlying the classical Gaussian prediction intervals, including linearity and homoskedasticity, are unlikely to hold exactly in practice. Consequently, the empirical analysis should primarily be interpreted as an investigation of the practical robustness and adaptivity of the different prediction interval methods in a realistic applied setting.

Evaluation Metrics

For a fixed realization of the training, calibration, and test split, the empirical coverage on the test set is computed as

$$\hat{C} = \frac{1}{n_{\text{test}}} \sum_{i \in I^{\text{test}}} \mathbb{1}\{Y_i \in C(X_i)\}.$$

As discussed in Sections 2.4.3 and 3.1, the empirical coverage $\hat{C}$ may be interpreted as a numerical approximation of the training-conditional coverage for the realized training split.

The average interval length is computed as

$$\hat{\mathcal{L}} = \frac{1}{n_{\text{test}}} \sum_{i \in I^{\text{test}}} (U(X_i) - L(X_i)),$$

where $L(X_i)$ and $U(X_i)$ denote the lower and upper bounds of the prediction interval.

Since all prediction intervals are constructed on the logarithmic price scale and subsequently transformed back to the original scale, uncertainty is expressed in relative rather than absolute terms on the original price scale.

For quantile regression forests, the interval width already varies across the covariate space on the logarithmic scale through the estimated conditional quantiles. Therefore, the exponential transformation further amplifies this local adaptivity on the original price scale. For the Gaussian and conformal prediction intervals, uncertainty is incorporated through a symmetric additive correction around the predicted log-price, $\hat{\mu}(x)$, either through the Gaussian prediction correction or through a global conformal quantile. In particular, prediction intervals of the form

$$[\hat{\mu}(x) - a(x), \hat{\mu}(x) + a(x)]$$

obtain interval widths

$$\exp(\hat{\mu}(x) + a(x)) - \exp(\hat{\mu}(x) - a(x)) = \exp(\hat{\mu}(x)) (\exp(a(x)) - \exp(-a(x))).$$

Consequently, interval widths on the original price scale increase proportionally with the predicted price level. Therefore, more expensive properties naturally receive wider prediction intervals even when the additive correction on the logarithmic scale remains similar across observations.

To reduce variability induced by random sample splitting, the entire procedure is repeated $B = 1000$ times using independent random splits of the data. The reported performance measures are then obtained by averaging across repetitions,

$$\bar{C} = \frac{1}{B} \sum_{b=1}^B \hat{C}^{(b)}, \quad \bar{\mathcal{L}} = \frac{1}{B} \sum_{b=1}^B \bar{\mathcal{L}}^{(b)}.$$

4.3 Results

4.3.1 Gaussian Prediction Intervals

Although the Gaussian prediction intervals do not provide distribution-free validity guarantees, the empirical marginal coverage remains close to the nominal target level. Across the test set, the empirical marginal coverage equals approximately 91.1%, indicating that the intervals are reasonably calibrated on average.

To further investigate the local behaviour of the Gaussian intervals, the empirical coverage and interval widths are analysed across municipalities, price levels, and sale years. The results are presented in Figure 11.

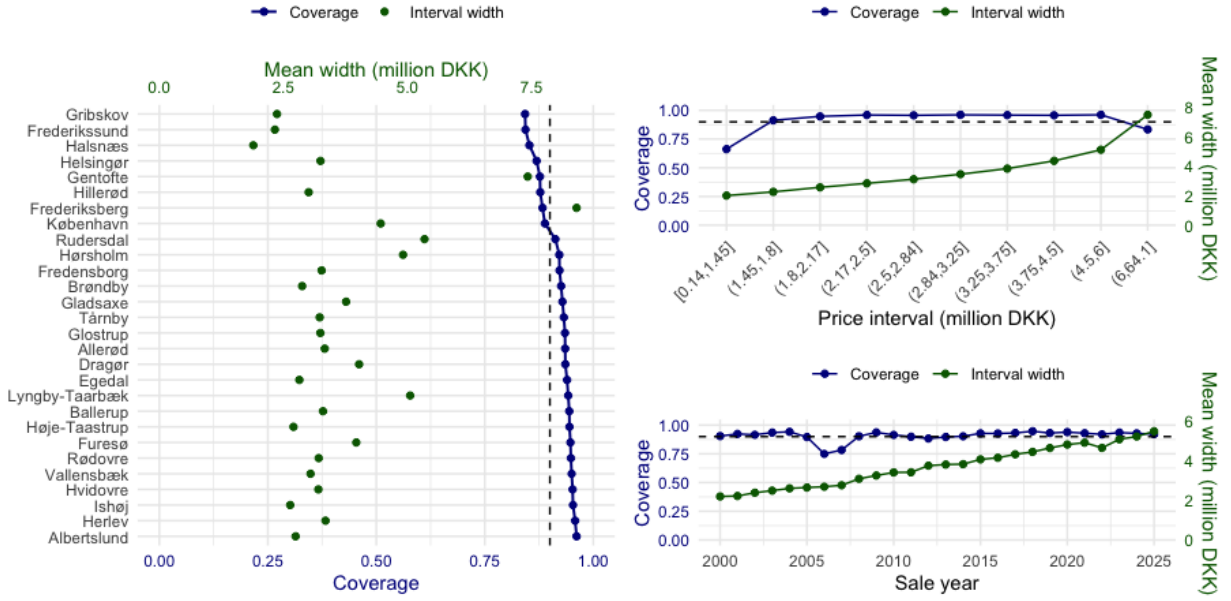


Figure 11: Empirical coverage and average prediction interval width for Gaussian prediction intervals across regions of the feature space. The dashed black lines indicate the nominal coverage level of 90%. **Left:** Coverage and interval width by municipality. **Top right:** Coverage and interval width across empirical price bins constructed from deciles of the observed transaction prices. **Bottom right:** Coverage and interval width by sale year.

Several important patterns emerge in Figure 11. First, despite achieving approximately valid marginal coverage overall, the intervals exhibit noticeable conditional variation in coverage across the feature space. In particular, clear undercoverage is observed for both low- and high-priced properties, whereas middle price segments exhibit small overcoverage. Similarly, noticeable variation in empirical coverage appears across municipalities, with some areas falling considerably below the nominal 90% level. These findings indicate that the Gaussian model assumptions are violated in practice, despite the overall marginal coverage remaining close to the desired target level.

Prediction interval widths increase strongly with both price level and sale year. This increase is induced by the exponential back-transformation from the logarithmic scale, as discussed in Section 4.2. Consequently, a constant-width interval on the logarithmic scale corresponds to an interval whose width increases proportionally with the predicted price level on the original scale. The geographical variation in interval width therefore largely reflects the variation in average housing prices across municipalities, shown in Figure 10, with more expensive municipalities naturally exhibiting wider prediction intervals.

A similar pattern is observed across sale years, where interval widths increase steadily over time. This behaviour is consistent with the increase in housing prices illustrated in Figure 10. Despite the changing market conditions over time, the empirical coverage remains relatively stable after the mid-2000s, suggesting that the Gaussian intervals still adapt reasonably well to the overall increase in market scale.

Overall, the results illustrate an important distinction between marginal and conditional validity. Even though the Gaussian prediction intervals achieve approximately valid marginal coverage, systematic overcoverage and undercoverage remain across different regions of the feature space.

4.3.2 Conformal Prediction Intervals

The limitations of the Gaussian prediction intervals naturally motivate the use of conformal prediction. Whereas the Gaussian intervals rely on strong assumptions regarding linearity, homoskedasticity, and Gaussian errors, split conformal prediction requires only exchangeability and provides finite-sample marginal validity independently of the underlying prediction model. It is therefore of particular interest to examine whether the conformal calibration step improves coverage behaviour in the heterogeneous housing market setting and how this affects interval efficiency.

We begin by examining the overall empirical behaviour of split conformal prediction across repeated random sample splits. The objective is both to assess whether the method achieves the desired marginal coverage level on average and to investigate the variability induced by the random train/calibration/test partitioning. From the perspective of training-conditional validity, each realized sample split induces a different fitted prediction model and a different conformal correction term. Consequently, the empirical coverage obtained on the corresponding test set may vary across repetitions, even though the method guarantees marginal validity on average over the random splits.

To investigate this behaviour empirically, the entire split conformal procedure is repeated $B = 1000$ times using independent stratified random splits of the data as described in Section 4.2. For each repetition, the empirical coverage and average prediction interval width are computed on the corresponding test set. The resulting distributions illustrate how the realized empirical coverage and interval widths vary across different training and calibration samples.

Figure 12 summarizes the empirical performance of split conformal prediction across the repeated sample splits.

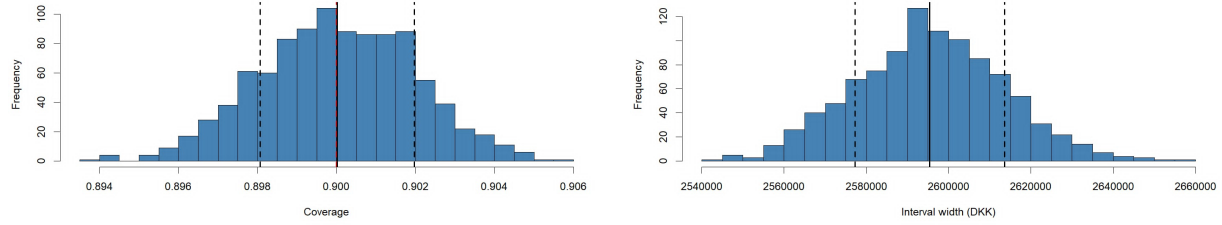


Figure 12: Left: Distribution of empirical coverage over 1000 repeated stratified sample splits. The red dashed line indicates the nominal coverage level $1 - \alpha = 0.9$, while the black dashed lines denote one standard deviation around the mean coverage. **Right:** Distribution of average prediction interval widths over 1000 repeated stratified sample splits. The black dashed lines denote one standard deviation around the mean interval width.

The empirical coverage is centered very close to the nominal target level of $1 - \alpha = 0.9$, indicating that the conformal intervals achieve stable marginal coverage across repeated random splits of the dataset. The variability across repetitions is relatively small, as reflected by the narrow spread around the mean coverage. From the perspective of training-conditional validity, this suggests that the realized coverage remains stable across different training and calibration samples.

The interval widths are likewise relatively stable across repetitions, although the intervals themselves remain fairly wide in absolute terms, with average widths around 2.6 million DKK. This reflects the variability and complexity of the housing market, where prices depend on many interacting structural, geographic, and temporal factors.

However, stable marginal behaviour across repeated sample splits does not imply that the empirical coverage remains uniform across different regions of the feature space. We therefore next investigate the local behaviour of split conformal prediction across different subpopulations of the data.

To investigate the local behaviour empirically, the conformal prediction intervals are analysed across several dimensions of the housing market, including sale year, municipality, and price level. As illustrated in Figure 10, the Copenhagen housing market exhibits temporal and geographical heterogeneity. Housing prices vary considerably across municipalities, increase strongly over time, and display pronounced right-skewness. These characteristics suggest that predictive uncertainty is unlikely to be homogeneous across the feature space.

Figure 13 summarizes the resulting empirical coverage and average interval widths across the different subgroups.

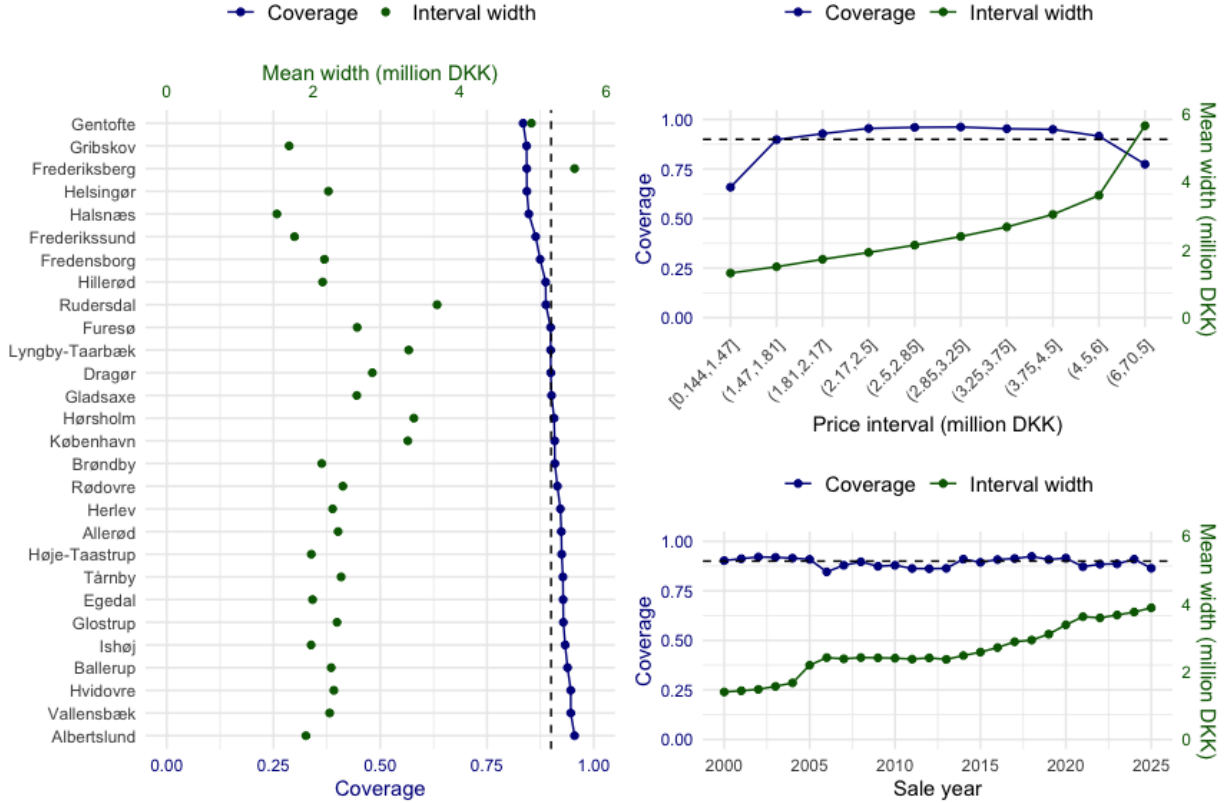


Figure 13: Empirical coverage and average prediction interval width for global conformal prediction intervals across regions of the feature space. The dashed black lines indicate the nominal coverage level of 90%. **Left:** Empirical coverage and average prediction interval width across municipalities. **Top right:** Empirical coverage and average prediction interval width across empirical price bins constructed from deciles of the observed transaction prices. **Bottom right:** Empirical coverage and average prediction interval width across sale years.

As illustrated in Figure 13, variation in local coverage is observed across both time and price level. Coverage fluctuates noticeably across sale years, with several periods exhibiting systematic undercoverage, particularly around 2006-2013 and again for the most recent years. At the same time, interval widths increase over time. This behaviour aligns closely with the increase in housing prices illustrated previously in Figure 10 together with the exponential back-transformation from the logarithmic scale.

A similarly pronounced pattern appears across empirical price bins. Low-priced properties exhibit undercoverage despite comparatively narrow intervals, whereas middle-priced properties tend to exhibit slight overcoverage. For the most expensive properties, the empirical coverage decreases again despite wider intervals. Nevertheless, the variation in empirical coverage across price levels indicates that the global conformal correction does not adequately adapt to local differences in predictive uncertainty across the price distribution.

Figure 13 furthermore reveals considerable geographical heterogeneity across municipalities.

Several municipalities exhibit empirical coverage below the target level, while others are noticeably overconservative. Expensive municipalities generally display wider intervals than lower-priced areas. This geographical variation aligns closely with the descriptive patterns observed previously in Figure 10.

Taken together, these results illustrate a key limitation of global conformal prediction. Although the method guarantees valid marginal coverage, a single global conformal quantile is insufficient to capture the local heterogeneity present in the housing market. This motivates localized conformal procedures, where calibration is performed separately within predefined subgroups of the feature space.

4.3.3 Mondrian Conformal Prediction

We next consider Mondrian conformal prediction, where conformal inference is performed separately within predefined subgroups of the feature space rather than globally across all observations. The calibration set is partitioned into disjoint groups

$$\mathcal{A}_1, \dots, \mathcal{A}_K,$$

and separate conformity quantiles are computed within each subgroup. Prediction intervals are then constructed using the conformity quantile corresponding to the appropriate group of the test observation.

A natural approach would be to partition the observations directly according to housing price level, since the previous analysis revealed differences in both empirical coverage and interval width across the price distribution. However, such a partitioning is problematic in practice because the housing price constitutes the response variable rather than an observed feature available at prediction time. Consequently, the Mondrian groups must instead be constructed using observable covariates.

In the present application, two different partitioning strategies are therefore considered: partitioning by sale year and partitioning by municipality groups. The temporal partitioning directly adapts the conformal inference to year-specific market conditions and volatility. For the geographical partitioning, municipalities are grouped according to average housing price level, thereby providing a proxy for local market characteristics and price-related heterogeneity. Expensive municipalities generally exhibit larger price variability and wider residual distributions than lower-priced municipalities. The municipality-based Mondrian procedure indirectly adapts the conformal inference to local differences in housing price level and predictive uncertainty.

Figure 14 summarizes the resulting empirical coverage and average interval widths.

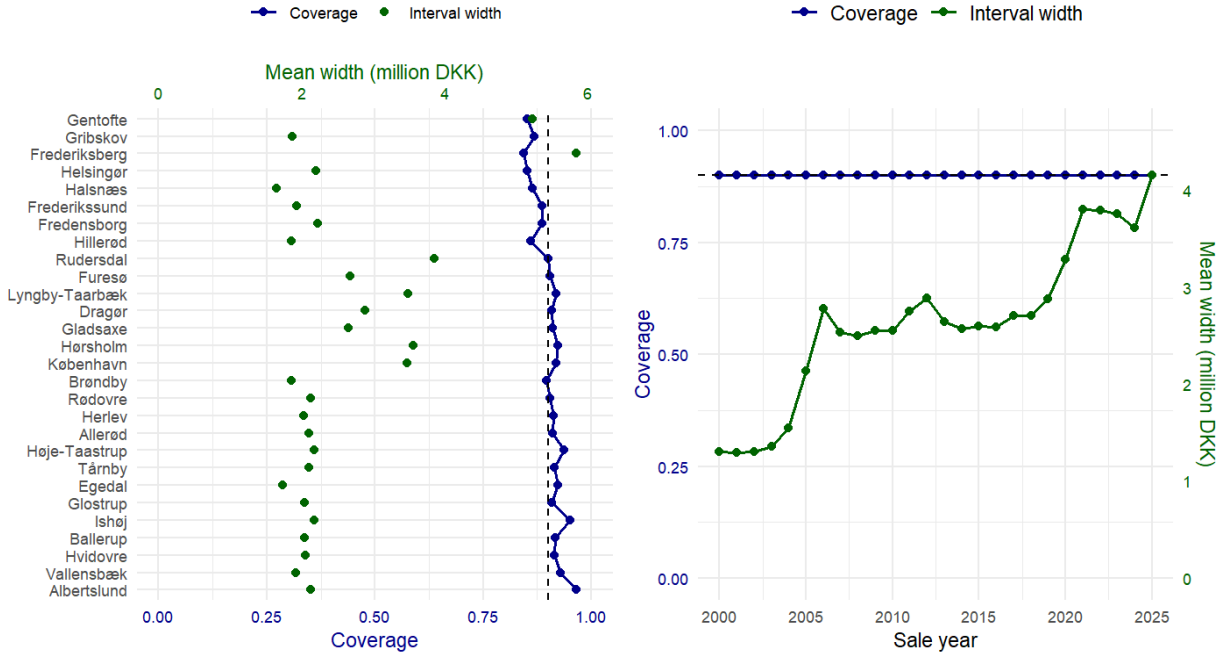


Figure 14: Empirical coverage and average prediction interval width for Mondrian conformal prediction intervals across regions of the feature space. The dashed black lines indicate the nominal coverage level of 90%. **Left:** Empirical coverage and average prediction interval width for municipality-based Mondrian conformal prediction using 11 municipality groups constructed according to average housing price level. **Right:** Empirical coverage and average prediction interval width for year-based Mondrian conformal prediction.

The year-based Mondrian procedure achieves stable empirical coverage across all years consistently close to the nominal target level of 90%. Compared to standard global conformal prediction, the temporal variation observed previously is eliminated. At the same time, interval widths continue to vary considerably across years, reflecting the scaling induced by the exponential back-transformation from the logarithmic scale. The municipality-based Mondrian procedure likewise reduces part of the geographical heterogeneity observed under global conformal prediction. Empirical coverage becomes more concentrated around the nominal level, although some municipality-specific variation remains. This remaining heterogeneity is expected, since the localized conformal quantiles are estimated only within relatively coarse municipality groups rather than separately for each municipality.

Overall, the results illustrate how localized conformal inference can partially adapt to structured heteroskedasticity in the housing market. While exact conditional validity is not achieved at the municipality level, the Mondrian procedure improves local stability relative to standard global conformal prediction. Increasing the number of Mondrian partitions may further improve local adaptivity and reduce subgroup-specific undercoverage. However, finer partitions also reduce the effective calibration sample size within each subgroup, leading to more variable conformity quantiles and potentially wider prediction intervals.

4.3.4 Quantile Regression

While conformal prediction constructs prediction intervals through calibration of global or localized nonconformity scores, quantile regression instead estimates the conditional quantiles of the response distribution directly. Consequently, the interval width adapts continuously across the feature space through the estimated lower and upper conditional quantiles rather than through a fixed conformal correction term.

Figure 15 summarizes the empirical coverage and average interval widths for quantile regression across municipalities, sale years, and price levels.

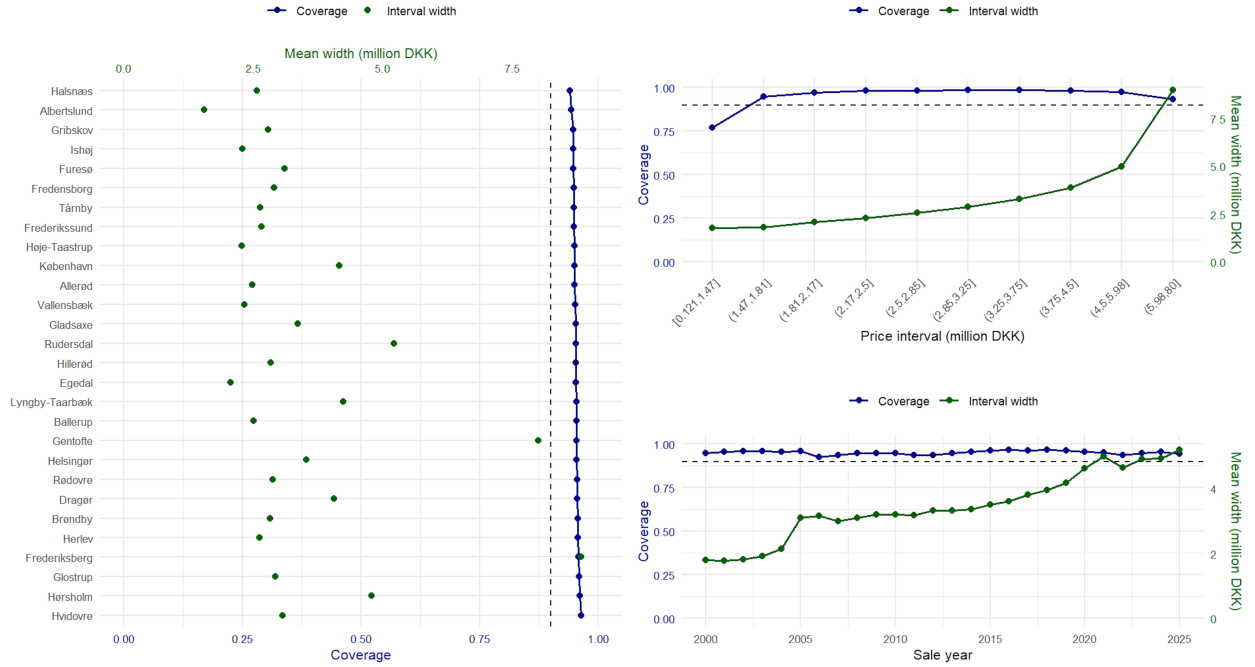


Figure 15: Empirical coverage and average prediction interval width for quantile regression prediction intervals across regions of the feature space. The dashed black lines indicate the nominal coverage level of 90%. **Left:** Empirical coverage and average prediction interval width across municipalities. **Top right:** Empirical coverage and average prediction interval width across empirical price bins constructed from deciles of the observed transaction prices. **Bottom right:** Empirical coverage and average prediction interval width across sale years.

Overall, the quantile regression intervals achieve empirical marginal coverage above the nominal target level of (90%). In fact, the average empirical coverage is approximately 95%, compared to roughly 90% for global conformal prediction. The empirical coverage furthermore appears considerably more stable across both sale years and municipalities compared to global conformal prediction. In particular, the temporal undercoverage previously observed during periods of elevated market volatility is largely eliminated.

However, the improved empirical coverage is accompanied by substantially wider prediction intervals with an average interval length of approximately DKK 3.5 million compared to

DKK 2.6 million for the global conformal prediction. While overcoverage may appear desirable from a validity perspective, it indicates that the intervals are conservative and may provide less informative uncertainty quantification in practice. The increased empirical coverage therefore comes at the expense of reduced efficiency.

The interval widths vary across the feature space through the estimated conditional quantiles. In particular, wider intervals are observed for more expensive properties and during periods with elevated housing prices, indicating heterogeneity in the conditional distribution of housing prices. Some conditional heterogeneity nevertheless remains. The lowest price segment still exhibits noticeable undercoverage, while several middle-price segments display moderate overcoverage. This suggests that although quantile regression captures an important part of the heteroskedastic structure of the housing market, the estimated conditional quantiles are not perfectly calibrated across the entire feature space.

The municipality-specific results indicate remarkably stable empirical coverage across geographical regions. Compared to both Gaussian prediction intervals and standard split conformal prediction, the municipality-specific variation observed previously is largely eliminated. This suggests that quantile regression successfully captures important local differences in the conditional distribution of housing prices and adapts naturally to geographical heterogeneity in the market.

Overall, the results demonstrate that quantile regression provides more adaptive prediction intervals than both standard split conformal prediction and Mondrian split conformal prediction. At the same time, the consistently elevated coverage levels suggest that this adaptivity is achieved through intervals that are often wider than necessary. The results therefore illustrate that local adaptivity alone does not guarantee efficient uncertainty quantification.

4.4 Discussion

The empirical analysis highlights several important aspects of predictive uncertainty quantification in the Copenhagen housing market. Across all methods considered, the results demonstrate that constructing valid and informative prediction intervals in highly heterogeneous and heteroskedastic settings remains challenging. In particular, the case study illustrates the fundamental distinction between marginal validity and stable local coverage behaviour across different regions of the feature space.

The classical Gaussian prediction intervals constructed from the linear regression model achieve empirical marginal coverage above the nominal target level of 90%. Nevertheless, local variation in coverage is observed across municipalities, years, and price levels. These findings suggest that several of the classical assumptions underlying the Gaussian linear regression model are violated in the housing market data. In particular, the relationship between housing prices and covariates appears strongly nonlinear, while the residual variability clearly depends on geography, price level, and time, contradicting the homoskedasticity assumption. Consequently, simple homoskedastic Gaussian prediction intervals are insufficient for obtaining stable local coverage behaviour despite performing well on average.

The global conformal prediction procedure likewise achieves empirical marginal validity across repeated random sample splits. Moreover, the method exhibits relatively stable behaviour with respect to the random train/calibration/test partitioning, indicating that the conformal inference procedure is robust in finite samples. However, some conditional heterogeneity remains across municipalities, years, and price levels. In particular, low-priced properties and periods of elevated market volatility exhibit systematic deviations from the nominal coverage level despite the overall marginal guarantee. These findings illustrate a key limitation of global conformal inference: although the procedure guarantees valid marginal coverage, a single global conformity quantile is insufficient to adapt fully to strong local variation in predictive uncertainty.

The localized Mondrian conformal procedures partially mitigate these issues. The year-based Mondrian calibration is particularly successful, producing highly stable empirical coverage across sale years. By calibrating separately within each year, the procedure effectively adapts to temporal changes in market volatility and housing prices. The municipality-based Mondrian procedure likewise reduces part of the geographical heterogeneity observed under global conformal prediction. Since the municipality groups are constructed according to average housing price level, the procedure indirectly incorporates both geographical and price-related heterogeneity. Nevertheless, some municipality-specific variation in coverage remains, illustrating the fundamental trade-off between localization and calibration sample size. Finer partitions improve local adaptivity but simultaneously reduce the number of calibration observations available within each subgroup, leading to more variable conformity quantiles and potentially wider prediction intervals.

At the same time, the exact exchangeability assumption underlying the finite-sample conformal validity guarantee is unlikely to hold perfectly in the housing market data. The strong temporal trends and changing market conditions observed across years suggest the presence of distributional shifts that may partially violate the exchangeability assumption in practice. The empirical results therefore illustrate both the robustness and the limitations of split conformal prediction under approximate rather than exact exchangeability.

Nevertheless, somewhat surprisingly, standard split conformal prediction achieves the narrowest intervals among all considered methods. Combined with its valid marginal coverage, this indicates that the method remains highly competitive despite its limited ability to adapt to local variation in predictive uncertainty.

Among the considered approaches, quantile regression produces the most locally adaptive prediction intervals. By estimating conditional quantiles directly, the method is able to accommodate heterogeneity across municipalities, sale years, and price levels. Empirically, the resulting intervals exhibit more stable local coverage behaviour than both Gaussian prediction intervals and standard split conformal prediction, with considerably less variation across subgroups of the feature space.

However, this improved local coverage performance comes at a cost in terms of interval width.

Although the average empirical coverage exceeds the nominal 90%-target and reaches approximately 95%, the corresponding prediction intervals are considerably wider than those obtained from global conformal prediction. Consequently, the increased local stability is achieved partly through conservative interval estimates, which raises the question of whether the additional empirical coverage is practically valuable given the associated loss of efficiency.

A possible explanation is the considerable tuning sensitivity of quantile regression forests. The quality of the estimated conditional quantiles depends strongly on hyperparameter choices. As discussed previously, substantial tuning was required in order to obtain satisfactory empirical coverage performance. The resulting intervals may therefore reflect a tendency toward conservative quantile estimation rather than accurate calibration of the conditional quantiles themselves.

Unlike conformal prediction, quantile regression furthermore does not provide finite-sample validity guarantees. Instead, its validity relies on accurate estimation of the conditional quantiles and is therefore inherently model-dependent. The strong empirical performance observed in the present case study may partly be explained by the large sample size available, which facilitates reliable estimation of complex conditional quantile functions.

The strong empirical performance of quantile regression also limits the practical gains obtained from conformalized quantile regression (CQR) introduced in Section 3.6. Since the quantile regression model already achieves empirical coverage well above the nominal ($1 - \alpha = 0.90$) level across most subgroups, the conformal correction term remains small, and the resulting CQR intervals are nearly identical to those of quantile regression forest. Consequently, a separate analysis of CQR is not presented in the case study.

An important practical aspect of the analysis is the logarithmic transformation of housing prices. The transformation is necessary due to the pronounced right-skewness of the response distribution and improves the stability of both regression and quantile estimation. However, prediction intervals constructed on the logarithmic scale are transformed back to the original price scale. Consequently, interval widths on the original scale increase proportionally with the predicted price level. Part of the large interval widths observed for expensive properties, therefore, reflects the exponential back-transformation itself rather than increased predictive uncertainty alone.

The analysis furthermore highlights an important practical limitation of split conformal prediction methods. Split conformal procedures require a separate calibration dataset in addition to the training set, implying that part of the available data cannot be used directly for model estimation. In the present analysis, the same training split is used across all methods for comparability. Consequently, the Gaussian and quantile regression approaches are fitted using only the training observations despite not requiring a separate calibration step. In practice, these methods could therefore potentially achieve even better predictive performance if the calibration observations were incorporated directly into the model estimation procedure.

Overall, the empirical findings illustrate the practical importance of balancing validity, adaptivity, and interval efficiency when constructing prediction intervals in complex real-world settings. While conformal prediction provides strong finite-sample marginal guarantees, additional localization or fully conditional modelling is often necessary in order to obtain stable uncertainty quantification across heterogeneous regions of the feature space.

From a practical perspective, the relative importance of validity, efficiency, and local adaptivity depends strongly on the intended use of the prediction intervals. For applications such as portfolio risk management, taxation, mortgage lending, or large-scale automated valuation systems, valid coverage guarantees may be of primary importance, making conformal prediction an attractive choice despite its limited local adaptivity. In contrast, applications that require accurate property-specific uncertainty assessments, such as individual property valuation, investment analysis, or decision support for buyers and sellers, may place greater emphasis on local adaptivity. In such settings, methods based on conditional quantile estimation may provide more informative uncertainty estimates, even at the cost of weaker theoretical guarantees and increased tuning complexity. The results of this case study suggest that the choice of prediction interval method should therefore be guided not only by overall empirical coverage performance, but also by the specific objectives and risk considerations of the application.

5 Conclusion

This thesis has investigated the construction of prediction intervals from both theoretical and empirical perspectives, with particular focus on the trade-offs between validity, efficiency, and local adaptivity. The analysis considered classical Gaussian prediction intervals, residual-based methods, quantile regression, split conformal prediction, Mondrian conformal prediction, and conformalized quantile regression. The methods were studied through theoretical analysis, simulation experiments, and a real-world case study involving housing prices in the Capital Region of Denmark. Overall, the thesis contributes a unified theoretical and empirical assessment of prediction interval methods, highlighting how different notions of validity, efficiency, and adaptivity interact in practice. The results demonstrate that methods with similar marginal validity properties may behave very differently locally, emphasizing the importance of evaluating predictive uncertainty beyond global performance measures.

Diving into the different parts of the thesis, the theoretical analysis revealed several fundamental distinctions between the considered approaches. Classical Gaussian linear prediction intervals provide exact finite-sample validity under strong parametric assumptions, including linearity, homoskedastic Gaussian errors, and correct model specification. Quantile regression methods relax many of these assumptions by directly estimating conditional quantiles and thereby naturally accommodating heterogeneous uncertainty. However, these methods rely on asymptotic arguments and model-dependent assumptions. In contrast, conformal prediction provides finite-sample distribution-free marginal validity under minimal assumptions, requiring only exchangeability of the observations. At the same time, the analysis demonstrated that exact conditional validity is generally unattainable in distribution-free settings, highlighting a fundamental limitation of conformal prediction methods.

The simulation experiments confirmed these trade-offs empirically. Standard split conformal prediction consistently achieved stable empirical marginal coverage close to the nominal target level across a wide range of experimental settings. However, the experiments also demonstrated that globally calibrated conformal intervals may fail to adapt efficiently to heterogeneous uncertainty across the covariate space. In particular, substantial conditional miscoverage was observed in settings involving heteroskedasticity, asymmetric noise, and nonlinear regression structure. Quantile regression methods exhibited more adaptive interval widths and achieved more stable local coverage behaviour across the feature space. In many settings, the methods successfully captured important heteroskedastic patterns in the conditional distribution. However, the simulation experiments also revealed that quantile regression may exhibit finite-sample undercoverage when the estimated conditional quantiles are inaccurate or insufficiently flexible.

The empirical case study on housing prices further revealed the practical challenges of uncertainty quantification in highly heterogeneous real-world settings. The empirical analysis exposed temporal, geographical, and price-dependent heterogeneity in predictive uncertainty. Global conformal prediction achieved stable empirical marginal coverage but displayed pronounced local variation in empirical coverage across municipalities, years, and price levels. The Gaussian prediction intervals exhibited similar behaviour, reflecting violations of the

underlying parametric assumptions. The localized Mondrian conformal procedures furthermore improved local stability relative to global conformal prediction by partially adapting the calibration procedure to local market structure. Quantile regression provided the most adaptive intervals and achieved remarkably stable empirical coverage across large parts of the feature space; however, at the cost of wide intervals. Moreover, the experiments also demonstrated the fundamental trade-off between localization and calibration sample size: finer partitions improve local adaptivity but simultaneously increase the variability of the estimated conformity quantiles.

The results demonstrate that no single prediction interval method can simultaneously achieve optimal validity, efficiency, and local adaptivity across all settings. Distribution-free conformal methods provide strong finite-sample guarantees but may struggle to adapt efficiently to local heterogeneity. Fully conditional approaches, such as quantile regression, achieve greater adaptivity but rely on weaker asymptotic guarantees and are highly sensitive to both model specification and tuning choices. Hence, the empirical findings highlight the importance of carefully balancing theoretical guarantees and practical adaptivity when constructing prediction intervals in complex predictive settings.

The relative importance of these properties ultimately depends on the intended application. In settings such as mortgage lending, taxation, insurance, and risk management, valid and well-calibrated prediction intervals may be more important than local adaptivity, since systematic undercoverage can have substantial financial consequences. Conversely, applications requiring highly individualized uncertainty assessments, such as property valuation, investment analysis, or decision support for buyers and sellers, may place greater emphasis on local adaptivity and property-specific uncertainty estimates. Thus, the results of this thesis suggest that the choice of prediction interval method should be guided not only by overall empirical coverage performance, but also by the specific objectives and risk considerations of the application at hand.

Several directions for future research naturally emerge from the present analysis. First, the comparison could be extended to a broader class of underlying predictive models, including gradient boosting methods, neural networks, and other modern ensemble approaches. Second, the analysis could be extended to high-dimensional settings where the number of covariates is large relative to the sample size and where both estimation accuracy and conformal inference become increasingly challenging. Third, future work could investigate alternative conformal methodologies beyond the split conformal framework considered in this thesis. Examples include jackknife+ and cross-conformal methods, which improve data efficiency, as well as localized and adaptive conformal procedures that seek to improve empirical coverage behavior in heterogeneous settings through covariate-dependent calibration (Barber et al., 2021; Angelopoulos et al., 2025; Lei and Wasserman, 2014).

Another important direction concerns the efficient use of data. In the present analysis, the same training split was used across methods for comparability. However, methods such as Gaussian prediction intervals and quantile regression do not require a separate calibration dataset and could therefore potentially benefit from utilizing both the training and calibra-

tion observations directly for model estimation. Future work could investigate the practical gains obtained from such approaches and compare them to the robustness provided by conformal prediction. More generally, future research could investigate alternative allocations between training and calibration observations within conformal prediction procedures. The choice of split proportion directly affects the trade-off between predictive model accuracy and the stability of the conformal calibration quantile. Larger training sets may improve the underlying predictive model, whereas larger calibration sets may yield more stable conformal corrections. Understanding this trade-off more systematically may therefore provide important practical guidance for conformal prediction in finite-sample settings.

Finally, although the present thesis focused exclusively on regression problems with continuous responses, many of the underlying ideas extend naturally to classification problems and other structured prediction tasks. More broadly, conformal prediction has recently attracted attention in medical diagnostics, image recognition, and risk-sensitive decision-making, where valid uncertainty quantification is often critically important (Balasubramanian et al., 2014; Lei and Wasserman, 2014).

Overall, the thesis demonstrates both the strengths and the limitations of modern prediction interval methods in heterogeneous predictive settings. While progress has been achieved in recent years, the construction of prediction intervals that are simultaneously valid, efficient, and locally adaptive remains a fundamentally challenging statistical problem.

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A Appendix

A.1 Tables

Table 2: Empirical coverage across the four experimental settings.

Experiment	Method	$n = 20$	$n = 80$	$n = 320$	$n = 1280$
Linear, Gaussian noise	Gaus (LM)	0.903	0.895	0.901	0.901
	CP (LM)	0.904	0.908	0.900	0.903
	CP (RF)	0.911	0.905	0.901	0.903
	QR (LM)	0.685	0.866	0.889	0.900
	QR (RF)	0.821	0.870	0.886	0.897
	RQ (LM)	0.760	0.864	0.890	0.900
	RQ (RF)	0.760	0.859	0.889	0.900
	CQR (RF)	0.871	0.889	0.892	0.896
Linear, Asymmetric noise	Gaus (LM)	0.901	0.921	0.925	0.929
	CP (LM)	0.903	0.900	0.903	0.902
	CP (RF)	0.904	0.899	0.901	0.901
	QR (LM)	0.670	0.858	0.889	0.897
	QR (RF)	0.816	0.877	0.897	0.911
	RQ (LM)	0.749	0.859	0.890	0.899
	RQ (RF)	0.738	0.853	0.890	0.898
	CQR (RF)	0.874	0.886	0.895	0.900
Non-linear, Gaussian noise	Gaus (LM)	0.910	0.899	0.901	0.900
	CP (RF)	0.907	0.900	0.903	0.899
	QR (RF)	0.790	0.834	0.881	0.891
	RQ (RF)	0.756	0.857	0.890	0.896
	CQR (RF)	0.894	0.883	0.892	0.899
Linear, heteroskedastic noise	Gaus (LM)	0.875	0.889	0.891	0.894
	CP (LM)	0.913	0.901	0.899	0.900
	CP (RF)	0.909	0.902	0.898	0.900
	QR (LM)	0.681	0.856	0.891	0.898
	QR (RF)	0.777	0.847	0.888	0.899
	RQ (LM)	0.755	0.858	0.887	0.897
	RQ (RF)	0.746	0.861	0.887	0.897
	CQR (RF)	0.875	0.887	0.897	0.900

Table 3: Relative interval lengths across the four experimental settings. Interval lengths are normalized relative to the oracle interval length.

Experiment	Method	$n = 20$	$n = 80$	$n = 320$	$n = 1280$
Linear, Gaussian noise	Gaus (LM)	1.23	1.03	1.01	1.01
	CP (LM)	1.23	1.08	1.01	1.01
	CP (RF)	1.79	1.16	1.05	1.02
	QR (LM)	0.76	1.01	0.99	1.01
	QR (RF)	1.41	1.19	1.05	1.03
	RQ (LM)	0.83	0.96	0.99	1.00
	RQ (RF)	1.21	1.03	1.01	1.02
	CQR (RF)	1.28	1.15	1.05	1.03
Linear, Asymmetric noise	Gaus (LM)	1.29	1.14	1.11	1.12
	CP (LM)	1.41	1.03	0.95	0.90
	CP (RF)	1.93	1.17	1.05	1.04
	QR (LM)	0.75	1.00	1.00	1.00
	QR (RF)	1.56	1.12	1.04	1.02
	RQ (LM)	0.82	0.94	0.99	1.00
	RQ (RF)	1.26	1.07	1.06	1.10
	CQR (RF)	1.47	1.15	1.01	1.00
Non-linear, Gaussian noise	Gaus (LM)	1.49	1.25	1.21	1.20
	CP (RF)	1.46	1.22	1.06	1.04
	QR (RF)	1.07	1.03	1.04	1.02
	RQ (RF)	0.97	1.09	1.03	1.03
	CQR (RF)	1.46	1.16	1.07	1.04
Linear, heteroskedastic noise	Gaus (LM)	1.34	1.18	1.16	1.16
	CP (LM)	1.67	1.27	1.20	1.19
	CP (RF)	1.98	1.34	1.23	1.25
	QR (LM)	0.79	0.98	1.00	1.00
	QR (RF)	1.15	1.02	1.04	1.02
	RQ (LM)	0.95	1.08	1.15	1.18
	RQ (RF)	1.26	1.15	1.19	1.23
	CQR (RF)	1.43	1.13	1.05	1.02

A.2 GAI Declaration

Please tick one of the following:

- ☒ I/we have used generative AI as an aid/tool
- ☐ I/we have **NOT** used generative AI as an aid/tool

List which GAI tools you have used and include the link to the platform (if possible):

- ChatGPT (<https://chatgpt.com>)
- Claude (<https://claude.ai>)

Describe how generative AI has been used in the exam paper:

GAI was used to:

- Generate ideas for data visualization and construction of plots in R
- Optimization of code
- Code debugging
- Proofreading
- English grammar
- Setting up references in Latex